# FLIP CLASSES AND ELEMENTARY EXPANSIONS OF VERTICAL-STRIP LLT POLYNOMIALS

PER ALEXANDERSSON AND LILAN DAI

Abstract. The vertical-strip LLT polynomials are, after the substitution $q \mapsto q+1$, sums of strict $P$-partition generating functions over orientations of decorated unit-interval graphs. A flip moves an ascending edge across a triangle while keeping its long edge. We show that the sum over every flip class is elementary-positive: it equals the sum of the elementary symmetric functions given by the highest reachable vertices of its orientations. This answers a question of Hwang, and it holds for circular unit-interval graphs as well. Summing over classes gives a new proof of the elementary expansion of Alexandersson and Sulzgruber, and its circular analog, which was open. We also obtain a flip-class refinement of the elementary expansion of LLT cumulants.

Contents

1. Introduction 1
2. Preliminaries 4
3. The four-term identity 9
4. Flip classes of ordinary graphs 11
5. Connecting orientations by flips 19
6. Flip classes of circular graphs 28
7. Consequences 32
References 34

## 1. Introduction

LLT polynomials were introduced by A. Lascoux, B. Leclerc and J.-Y. Thibon as $q$-deformations of products of Schur functions [LLT97]. I. Grojnowski and M. Haiman proved that they are Schur-positive [GH06]. LLT polynomials are the building blocks of the combinatorial formula for Macdonald polynomials of J. Haglund, M. Haiman and N. Loehr [HHL05a], and of the Shuffle Theorem, conjectured in [HHL$^+$05b] and proved by E. Carlsson and A. Mellit [CM17]; see also [Hag07].

The vertical-strip LLT polynomials form a subfamily which can be described by colorings of decorated unit-interval graphs [AP18]: if $\Gamma$ is such a graph on $[n]$, then

$$\mathrm{LLT}_\Gamma(\mathbf{x};q) = \sum_\kappa q^{\mathrm{asc}(\kappa)} x_{\kappa(1)} x_{\kappa(2)} \cdots x_{\kappa(n)},$$

2020 *Mathematics Subject Classification.* Primary 05E05; Secondary 05A19, 05C20.
*Key words and phrases.* LLT polynomials, elementary symmetric functions, unit-interval graphs, orientations, flips.

where $\kappa$ ranges over the colorings which increase strictly along the strict edges, and $\mathrm{asc}(\kappa)$ counts the other edges along which $\kappa$ increases. Without strict edges, they are the unicellular LLT polynomials, which are related by a plethystic substitution to the chromatic quasisymmetric functions of J. Shareshian and M. Wachs [SW12, SW16], see [CM17, GP16]. For chromatic symmetric functions, elementary positivity is the content of the Stanley–Stembridge conjecture [SS93, Sta95]. M. Guay-Paquet reduced it to unit-interval graphs [GP13], and T. Hikita proved it [Hik24]. The refinement of Shareshian and Wachs to general $q$ [SW16] remains open.

The first named author and R. Sulzgruber proved that, after the substitution $q \mapsto q+1$, the vertical-strip LLT polynomials expand positively in the elementary symmetric functions [AS22], as conjectured in [GHQR19, Ale20]; the positivity was obtained independently by M. D'Adderio [D'A20]. The expansion is

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = \sum_\theta q^{\mathrm{asc}(\theta)} \mathrm{e}_{\lambda(\theta)}(\mathbf{x}). \tag{1}$$

Here $\theta$ ranges over the orientations of the normal edges of $\Gamma$, $\mathrm{asc}(\theta)$ is the number of ascending edges, those oriented from the smaller to the larger vertex, and $\lambda(\theta)$ records the sizes of the sets of vertices with the same highest reachable vertex. The proof in [AS22] shows that both sides of (1) satisfy a system of linear relations which determines them. Local linear relations of this kind go back to the modular law of M. Guay-Paquet [GP13] and to the relations of S. J. Lee [Lee18]; see also [AN20]. The analogous question for circular unit-interval graphs, whose chromatic quasisymmetric functions were studied by B. Ellzey [Ell17a, Ell17b], was left open in [AS22, Sec. 6.8].

The same polynomial has a simpler expansion. Expanding $(1+q)^{\mathrm{asc}(\kappa)}$ shows that

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = \sum_\theta q^{\mathrm{asc}(\theta)} \mathrm{K}_\theta(\mathbf{x}), \tag{2}$$

where $P_+(\theta)$ is the partial order generated by the strict edges and the ascending edges of $\theta$, and $\mathrm{K}_P$ is the generating function of strict $P$-partitions. The summands $\mathrm{K}_\theta$ are not symmetric functions. The situation is the same for chromatic symmetric functions, which are sums of strict $P$-partition generating functions over the acyclic orientations of a graph [Sta95]. B.-H. Hwang introduced flips of acyclic orientations, which reverse a fork of two covers, and showed that the corresponding sums over flip classes are symmetric [Hwa24, Thm. 3.8] and Schur-positive. He conjectured that they are elementary-positive [Hwa24, Conj. 3.10]. He also mentioned an analogous flip on orientations for LLT polynomials, suggested by the first named author, and asked whether its class sums are Schur-positive [Hwa24, Sec. 5.3]. This is the flip of the present paper.

**Main results.** Let $u < v < w$ and let $\theta$ be an orientation in which $uv$ is ascending and $vw$ is not. If $uv$ and $uw$ are covers of $P_+(\theta)$, and $vw$ and $uw$ are covers after reversing $uv$ and $vw$, we say that the two orientations differ by a *flip*, see Definition 2.5. Thus a flip keeps a long edge $uw$ and moves an ascending short edge across the triangle. A flip preserves $\mathrm{asc}(\theta)$, and it changes $\mathrm{K}_\theta$ as well as $\lambda(\theta)$.

**Theorem A** (Theorem 2.11)**.** Let $\Gamma$ be an ordinary or circular decorated unit-interval graph, and let $C$ be a flip class of orientations of $\Gamma$. Then

$$\sum_{\theta \in C} \mathrm{K}_\theta(\mathbf{x}) = \sum_{\theta \in C} \mathrm{e}_{\lambda(\theta)}(\mathbf{x}).$$

For example, in the complete graph on three vertices the two orientations with ascending edges $\{12, 13\}$ and $\{13, 23\}$ form a flip class, and

$$\mathrm{K}_{\{12,13\}} + \mathrm{K}_{\{13,23\}} = \mathrm{e}_{21} + \mathrm{e}_3,$$

see Example 2.7. Thus every class sum is elementary-positive, and in particular Schur-positive, which answers the question of Hwang mentioned above. Summing Theorem A over all flip classes and comparing (2) with (1) we obtain the following.

**Theorem B** (Corollary 2.12)**.** The expansion (1) holds for every circular decorated unit-interval graph. Here $\theta$ ranges over the admissible orientations, in which the strict and ascending edges form no directed cycle, and highest reachable vertices are computed on the lift to $\mathbb{Z}$.

For ordinary graphs this is (1) itself, and our proof does not use the recursion of [AS22]. Theorem A refines (1): every flip class contributes an elementary-positive summand $q^{\mathrm{asc}(C)} \sum_{\theta \in C} \mathrm{e}_{\lambda(\theta)}$. The same refinement applies to the LLT cumulants of M. Dołęga and M. Kowalski [DK22].

**Corollary C** (Corollary 7.2)**.** The LLT cumulants of circular decorated unit-interval graphs, after $q \mapsto q + 1$, are sums of elementary-positive contributions of flip classes.

**Method.** The proof of Theorem A for ordinary graphs uses one local identity: for three vertices $u < v < w$ and any fixed set of other edges containing $uw$, the four ways of adding the short edges $uv$ and $vw$ satisfy

$$\Psi(\text{with } uv) + \Psi(\text{with } vw) = \Psi(\text{with neither}) + \Psi(\text{with both}),$$

both for $\Psi = \mathrm{K}$ and for $\Psi = \mathrm{e}_\lambda$, see Lemma 3.3. Summed over all orientations with a fixed number of edges, it gives refined forms of the two basic bounce relations of [AS22]. To pass from all orientations to a single flip class, we prove a connectivity theorem: in a graph without articulations, all connected orientations with a given number of ascending edges in which every corner is a cover form one flip class. Every other flip class of that size is expressed, with the same coefficients for both sides, in terms of flip classes of graphs with fewer edges. A graph with articulations is cut at the end of its first block, and the relation is transported to the whole graph by fixing the color of the cut vertex. The circular case is reduced to the ordinary one: a finite signed expansion writes every admissible orientation in terms of orientations whose edge lengths are differences of a potential, the coefficients are constant on flip classes, and the flip classes of such orientations can be drawn on a line.

Theorem A for ordinary and circular graphs, and the elementary expansions obtained from it, have been formally verified in the Lean proof assistant using Mathlib; the formalization is available at `https://github.com/PerAlexandersson/VerticalStripLLT`.

**Outline.** Section 2 contains the definitions and the precise statements. Section 3 proves the four-term identity. Section 4 proves Theorem A for ordinary graphs, assuming the connectivity theorem, which is proved in Section 5. Section 6 treats circular graphs, and Section 7 derives the elementary expansion and the result on cumulants, and ends with two questions.

## 2. Preliminaries

2.1. **Symmetric functions and $P$-partitions.** We use symmetric functions in $\mathbf{x} = (x_1, x_2, \dots)$ with coefficients in $\mathbb{Z}[q]$, see [Sta72] for the background on $P$-partitions. For an integer partition $\lambda$ we write $\mathrm{e}_\lambda = \mathrm{e}_{\lambda_1}\mathrm{e}_{\lambda_2}\cdots$. For a map $\kappa$ from a finite set $V$ to $\mathbb{N}^+$ we put $x_\kappa := \prod_{u\in V} x_{\kappa(u)}$.

Let $P$ be a partial order on a finite set $V$. A map $\kappa : V \to \mathbb{N}^+$ is a *strict $P$-partition* if $\kappa(u) < \kappa(v)$ whenever $u <_P v$, and we set

$$\mathrm{K}_P(\mathbf{x}) := \sum_\kappa x_\kappa,$$

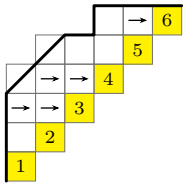
the sum ranging over all strict $P$-partitions. This is a quasisymmetric function, and it is symmetric when $P$ is a disjoint union of chains: a chain with $k$ elements contributes $\mathrm{e}_k$. We write $u \lessdot_P v$ when $v$ *covers* $u$ in $P$. If a set of directed edges generates a partial order $P$, then $\mathrm{K}_P$ counts the maps that increase strictly along every edge, and every cover of $P$ is one of the edges.

2.2. **Decorated unit-interval graphs.** An *area sequence* is a tuple $\mathbf{a} = (a_1, \dots, a_n)$ of nonnegative integers with $a_1 = 0$ and $a_{i+1} \le a_i + 1$. We put $b(i) := i - a_i$, so that $b$ is weakly increasing. A pair $uv$ with $u < v$ is a *natural edge* if $b(v) \le u$. The natural edges form a unit-interval graph: if $uw$ is a natural edge and $u < v < w$, then $uv$ and $vw$ are natural edges. There are $a_v$ natural edges ending at $v$, so $\sum_i a_i$ natural edges in total.

A natural edge $uv$ is a *corner* if no other natural edge $u'v'$ satisfies $u' \le u$ and $v \le v'$. Equivalently, $u = b(v)$, and either $v = n$ or $b(v+1) > b(v)$. Every natural edge lies inside a corner, and distinct corners have distinct tails and distinct heads. A *decorated unit-interval graph* $\Gamma = (\mathbf{a}, S)$ is an area sequence together with a set $S$ of corners, the *strict edges*. The other natural edges are *normal*. Decorated unit-interval graphs are the graphs $\Gamma_P$ of Schröder paths $P$ in [AS22, Sec. 2.6]: the natural edges are the cells below the Dyck path obtained by replacing every diagonal step by a north step followed by an east step, and the strict edges are the cells of the diagonal steps.

We draw $\Gamma$ as in [AS22]. The diagonal box in row $v$ is the vertex $v$, and the cell in column $u$ and row $v$, for $u < v$, is the pair $uv$. The natural edges are the cells between the Schröder path and the diagonal, and the diagonal steps of the path cross the strict edges.

**Example 2.1.** The Schröder path `nnnddeneee` of [AS22, Ex. 2.8] gives the area sequence $\mathbf{a} = (0, 1, 2, 3, 3, 2)$ with strict edges 14 and 25. The corners are 14, 25 and 46; only the first two are strict. In the diagram below, arrows mark the ascending edges of an orientation $\theta$, defined in the next subsection:




2.3. **Colorings and orientations.** A *coloring* of $\Gamma$ is a map $\kappa : [n] \to \mathbb{N}^+$ with $\kappa(u) < \kappa(v)$ for every strict edge $uv$. An *ascent* of $\kappa$ is a normal edge $uv$ with

$\kappa(u) < \kappa(v)$, and $\operatorname{asc}(\kappa)$ is the number of ascents. The *vertical-strip LLT polynomial* is

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q) \coloneqq \sum_\kappa q^{\operatorname{asc}(\kappa)} x_\kappa,$$

the sum ranging over all colorings of $\Gamma$. It is a symmetric function [AP18, HHL05a], and every vertical-strip LLT polynomial of [LLT97] arises in this way [AP18, CM17].

An *orientation* $\theta$ of $\Gamma$ directs every normal edge. A normal edge $uv$ with $u < v$ is *ascending* in $\theta$ if it is directed from $u$ to $v$, and $\operatorname{asc}(\theta)$ is the number of ascending edges. Strict edges are always directed upwards. We write $\mathcal{O}(\Gamma)$ for the set of orientations, and we identify $\theta$ with its set of ascending edges. We regard the set $\theta \cup S$ of ascending and strict edges as a directed graph on $[n]$. All its edges point upwards, so its transitive closure is a partial order $P_+(\theta)$ on $[n]$, and we write $\mathrm{K}_\theta \coloneqq \mathrm{K}_{P_+(\theta)}$.

**Lemma 2.2.** *For every decorated unit-interval graph* $\Gamma$*,*

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = \sum_{\theta \in \mathcal{O}(\Gamma)} q^{\operatorname{asc}(\theta)}\, \mathrm{K}_\theta(\mathbf{x}).$$

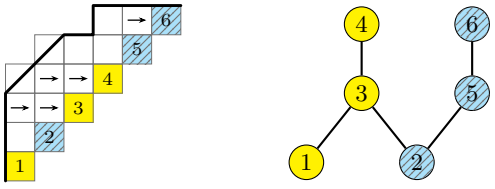
*Proof.* For a coloring $\kappa$, expand $(q+1)^{\operatorname{asc}(\kappa)}$ as a sum of $q^{|A|}$ over the subsets $A$ of the ascents of $\kappa$. A pair $(\kappa, A)$ is the same as an orientation $\theta = A$ together with a strict $P_+(\theta)$-partition $\kappa$, since $\kappa$ increases on the strict edges and on the edges of $A$. The claim follows. ∎

For a set $\eta$ of pairs $uv$ with $u < v$, regarded as edges directed upwards, the *highest reachable vertex* $\operatorname{hrv}(\eta, u)$ of a vertex $u$ is the largest vertex which can be reached from $u$ along edges of $\eta$. For an orientation $\theta$ we write $\operatorname{hrv}(\theta, u) \coloneqq \operatorname{hrv}(\theta \cup S, u)$, the largest $v$ with $u \le v$ in $P_+(\theta)$. The fibres of $\operatorname{hrv}(\theta, \cdot)$ form a set partition of $[n]$, and $\lambda(\theta)$ is the partition given by their sizes.

**Example 2.3.** We compute the highest reachable vertices of the orientation $\theta$ of Example 2.1. The graph $\theta \cup S$ has the strict edges 14, 25 and the ascending edges 13, 23, 24, 34, 56. Following its edges upwards, the vertices reachable from each vertex, and the largest among them, are

| $u$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| reachable from $u$ | $1, 3, 4$ | $2, 3, 4, 5, 6$ | $3, 4$ | 4 | $5, 6$ | 6 |
| $\operatorname{hrv}(\theta, u)$ | 4 | 6 | 4 | 4 | 6 | 6 |

Note that $\operatorname{hrv}(\theta, 2) = 6$ although the edges 23 and 24 lead to vertices whose highest reachable vertex is 4: the maximum is taken over all directed paths. The fibres are $\{1, 3, 4\}$ and $\{2, 5, 6\}$, so $\lambda(\theta) = (3, 3)$, and $\theta$ contributes $q^5 \mathrm{e}_{33}$ to Theorem 2.4 below, see also [AS22, Ex. 2.8]. Below, the diagonal boxes and the vertices are shaded by fibre, and the right picture is the Hasse diagram of $P_+(\theta)$:




The vertex 2 lies below both 4 and 6, and $\operatorname{hrv}(\theta, 2)$ is the larger of the two. The fibres need not be chains of $P_+(\theta)$, see Example 2.9, and $\mathrm{K}_\theta$ is not symmetric.

The following expansion was conjectured in [GHQR19, Ale20]. Its positivity was obtained independently by M. D'Adderio [D'A20].

**Theorem 2.4** ([AS22, Cor. 2.10])**.** *For every decorated unit-interval graph* $\Gamma$*,*

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = \sum_{\theta \in \mathcal{O}(\Gamma)} q^{\mathrm{asc}(\theta)}\, \mathrm{e}_{\lambda(\theta)}(\mathbf{x}).$$

Comparing with Lemma 2.2, the function $\mathrm{K}_\theta$ and the elementary function $\mathrm{e}_{\lambda(\theta)}$ agree after summing over all orientations with a given number of ascending edges, although they differ for individual orientations. The purpose of this paper is to identify much smaller sets of orientations over which they agree.

### 2.4. **Flips.**

**Definition 2.5.** Let $u < v < w$ be vertices such that $uw$ is a natural edge, and let $\theta$ be an orientation in which $uv$ is ascending and $vw$ is not. Let $\theta'$ be obtained from $\theta$ by reversing both $uv$ and $vw$, so that $vw$ is ascending in $\theta'$ and $uv$ is not. We say that $\theta$ and $\theta'$ differ by a *flip* if

$$u \lessdot v \text{ and } u \lessdot w \text{ in } P_+(\theta), \qquad \text{and} \qquad v \lessdot w \text{ and } u \lessdot w \text{ in } P_+(\theta').$$

The *flip classes* of $\Gamma$ are the equivalence classes of $\mathcal{O}(\Gamma)$ under the relation generated by flips, see Figure 1.

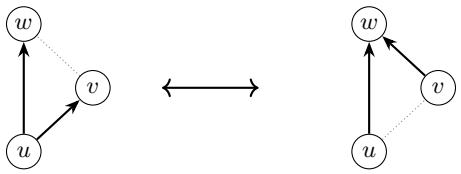



FIGURE 1. A flip at $u < v < w$, from $\theta$ (left) to $\theta'$ (right). The long edge $uw$ is kept, and the upward short edge moves from $uv$ to $vw$; dotted pairs are not upward edges. Both drawn edges on each side must be covers, and all other edges of $\theta \cup S$ are unchanged.

Since $u \lessdot w$ in $P_+(\theta)$, the pair $uw$ is an edge of $\theta \cup S$, the *long edge* of the flip. It is retained, and so are all other edges of $\theta \cup S$. The two exchanged *short edges* $uv$ and $vw$ lie strictly inside $uw$; hence they are not corners, and in particular they are normal. The definition is symmetric in $\theta$ and $\theta'$, with the roles of the two short edges exchanged. It is the LLT counterpart of the flips introduced by B.-H. Hwang for acyclic orientations [Hwa24], which reverse a fork of two covers.

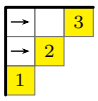
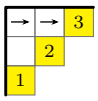
In an ordinary graph, two of the four cover conditions are automatic.

**Lemma 2.6.** *Let* $\Gamma$ *be ordinary, let* $u < v < w$ *with* $uw$ *natural, and let* $\theta$ *be an orientation in which* $uv$ *and* $uw$ *are covers of* $P_+(\theta)$*. Then* $\theta$ *and* $\theta - uv + vw$ *differ by a flip. Similarly, if* $vw$ *and* $uw$ *are covers of* $P_+(\theta)$*, then* $\theta$ *and* $\theta - vw + uv$ *differ by a flip.*

*Proof.* For the first claim, $vw \notin \theta$ since $uw$ is a cover; put $\theta' = \theta - uv + vw$. A path from $v$ to $w$ in $\theta' \cup S$ other than the edge $vw$ does not use $uv$, so it is a path in $\theta \cup S$, and preceded by $uv$ it would bypass $uw$. A path of length at least two from $u$ to $w$ in $\theta' \cup S$ either avoids $vw$, and then bypasses $uw$ in $\theta \cup S$, or consists of a

path from $u$ to $v$ followed by $vw$; that path from $u$ to $v$ avoids $uv$ and $vw$, so it bypasses $uv$ in $\theta \cup S$. Hence $v \lessdot w$ and $u \lessdot w$ in $P_+(\theta')$. The second claim follows in the same way. ■

**Example 2.7.** Let $\Gamma$ be the complete graph on $[3]$, with Schröder path `nnneee`. The two orientations

$\theta =$  and $\theta' =$ 

differ by a flip at $(1,2,3)$: the covers of $P_+(\theta)$ are $1 \lessdot 2$ and $1 \lessdot 3$, and those of $P_+(\theta')$ are $1 \lessdot 3$ and $2 \lessdot 3$. No other flip applies to them, so $C = \{\theta, \theta'\}$ is a flip class. The vertex 2 is its own highest reachable vertex in $\theta$, while all vertices reach 3 in $\theta'$. Hence

$$\mathrm{K}_\theta + \mathrm{K}_{\theta'} = \mathrm{e}_{21} + \mathrm{e}_3,$$

which can be checked on the coefficients of $x_1x_2x_3$ and $x_1^2x_2$. Neither summand on the left is symmetric.

**Lemma 2.8.** *On a flip class, the number of ascending edges, the connected components of the underlying undirected graph of $\theta \cup S$, and the set of corners which are edges of $\theta \cup S$ are constant.*

*Proof.* A flip exchanges one ascending edge for another and retains the long edge $uw$. Both short edges join $v$ to the component containing $u$ and $w$. A corner is not contained in a larger natural edge, so it is never a short edge of a flip. ■

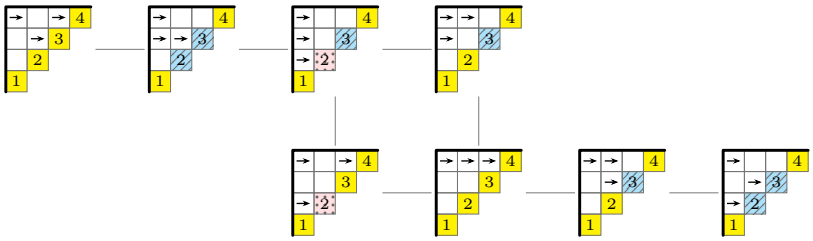


FIGURE 2. A flip class of the complete graph on $[4]$, with eight orientations, each with three ascending edges. Lines are flips, and the diagonal boxes are shaded by the fibres of $\mathrm{hrv}(\theta, \cdot)$.

**Example 2.9.** Let $\Gamma$ be the complete graph on $[4]$ without strict edges. The eight orientations in Figure 2 form a flip class $C$: every flip from one of them leads to another one, and the lines show that they are connected. Reading off the fibres,

$$\mathsf{E}(C) = \mathrm{e}_{211} + 2\mathrm{e}_{22} + 3\mathrm{e}_{31} + 2\mathrm{e}_4,$$

and by Theorem 2.11 this is also $\mathsf{K}(C)$. In the orientation with ascending edges 13, 14 and 24, the fibre $\{1,2,4\}$ is not a chain of $P_+(\theta)$, since 1 and 2 are incomparable. The 64 orientations of this graph form 39 flip classes.

Flip classes can be much larger. The following table counts the flip classes of all unit-interval graphs on $n$ vertices without strict edges; the largest classes occur for the complete graphs.

| $n$ | graphs | orientations | flip classes | classes of size $> 1$ | largest class |
|---|---|---|---|---|---|
| 3 | 5 | 17 | 16 | 1 | 2 |
| 4 | 14 | 171 | 132 | 25 | 8 |
| 5 | 42 | 3113 | 1559 | 522 | 84 |
| 6 | 132 | 106419 | 25276 | 11331 | 2589 |

2.5. **Circular graphs.** A *circular area sequence* is a tuple $\mathbf{a} = (a_1, \dots, a_n)$ with $n \geq 1$, $0 \leq a_i \leq n-1$ and $a_{i+1} \leq a_i + 1$ for all $i$, with indices read modulo $n$. We extend $\mathbf{a}$ periodically to all of $\mathbb{Z}$ and put $b(i) = i - a_i$, so that $b$ is weakly increasing and $b(i+n) = b(i) + n$. The *lifted natural edges* are the pairs $uv$ of integers with $b(v) \leq u < v$. They are invariant under translation by $n$, and their classes modulo $n$ are the *natural edges* of the *circular unit-interval graph*. A natural edge from the residue of $u$ to that of $v$ has length $v - u < n$; two residues may be joined by natural edges in both directions, and we regard these as different edges. Corners and strict edges are defined on the lift as before, and give a *circular decorated unit-interval graph* $\Gamma = (\mathbf{a}, S)$. An area sequence with $a_1 = 0$ has no natural edge ending at 1, and then $a_v \leq v - 1$ for all $v$; thus the ordinary graphs are precisely the circular graphs in which no lifted natural edge crosses from one period to the next. Circular unit-interval graphs and their chromatic quasisymmetric functions were studied in [Ell17a, Ell17b, AP18].

Colorings $\kappa : \mathbb{Z}/n\mathbb{Z} \to \mathbb{N}^+$, ascents and $\mathrm{LLT}_\Gamma(\mathbf{x}; q)$ are defined as before, with one ascent for every normal edge whose tail has a smaller color than its head. An orientation $\theta$ is again a set of ascending normal edges, and $\theta \cup S$ is the directed multigraph on $\mathbb{Z}/n\mathbb{Z}$ with the strict and ascending edges. Now $\theta \cup S$ may contain directed cycles, for example around the whole circle. We call $\theta$ *admissible* if $\theta \cup S$ is acyclic, and we write $\mathcal{O}(\Gamma)$ for the set of admissible orientations. A directed cycle admits no strictly increasing coloring, so Lemma 2.2 holds verbatim, with the sum over admissible orientations.

Let $\theta$ be admissible and let $\widetilde{\theta}$ be the periodic graph on $\mathbb{Z}$ consisting of the lifted strict and ascending edges. Its edges point upwards, so reachability in $\widetilde{\theta}$ is a partial order on $\mathbb{Z}$, and we take covers in this order. A path in $\widetilde{\theta}$ cannot visit two lifts of the same residue, since its projection would contain a directed cycle. Hence every integer $z$ has a largest reachable integer $\widetilde{\mathrm{hrv}}(\theta, z)$, and $\widetilde{\mathrm{hrv}}(\theta, z+n) = \widetilde{\mathrm{hrv}}(\theta, z) + n$. The highest reachable vertex $\mathrm{hrv}(\theta, u)$ is the residue of $\widetilde{\mathrm{hrv}}(\theta, u)$, and $\lambda(\theta)$ is the partition of the fibre sizes of $\mathrm{hrv}(\theta, \cdot)$ as before. For ordinary graphs these definitions agree with the previous ones.

**Example 2.10.** Let $\mathbf{a} = (1, 1, 1)$ with no strict edges. The natural edges form the directed cycle $1 \to 2 \to 3 \to 1$. The orientation with ascending edges $3 \to 1$ and $1 \to 2$ lifts to the path $3 \to 4 \to 5$, so all three vertices have highest reachable vertex 2, see Figure 3. The admissible orientations are the empty one, the three single edges and the three paths of two edges, and the right-hand side of Corollary 2.12 below is $\mathrm{e}_{111} + 3q\,\mathrm{e}_{21} + 3q^2\mathrm{e}_3$.

Flips are defined on the lift. Let $u < v < w < u + n$ be integers such that $uw$ is a lifted natural edge, and let $\theta, \theta'$ be admissible orientations related as in Definition 2.5, exchanging the residue classes of $uv$ and $vw$. They differ by a *flip* if the four cover conditions of Definition 2.5 hold in the reachability orders of $\widetilde{\theta}$ and $\widetilde{\theta'}$. Lemma 2.8 holds for circular graphs as well, with the same proof.

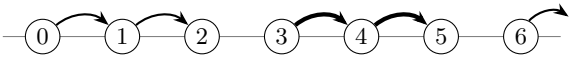



FIGURE 3. The lift to $\mathbb{Z}$ of the orientation in Example 2.10 with ascending edges $3 \to 1$ and $1 \to 2$; the integer $z$ lies over the residue of $z$ modulo 3. The path $3 \to 4 \to 5$ shows that 3 reaches the lift 5 of 2.

2.6. **Main results.** For a set $\Omega$ of orientations of $\Gamma$ we write

$$\mathsf{K}(\Omega) \coloneqq \sum_{\theta\in\Omega} \mathrm{K}_\theta(\mathbf{x}), \qquad \mathsf{E}(\Omega) \coloneqq \sum_{\theta\in\Omega} \mathrm{e}_{\lambda(\theta)}(\mathbf{x}).$$

**Theorem 2.11.** *Let $\Gamma$ be an ordinary or circular decorated unit-interval graph. Then $\mathsf{K}(C) = \mathsf{E}(C)$ for every flip class $C$ of $\Gamma$.*

The number of ascending edges is constant on $C$ by Lemma 2.8; we denote it by $\mathrm{asc}(C)$. Combining Theorem 2.11 with Lemma 2.2, every flip class contributes

$$q^{\mathrm{asc}(C)}\mathsf{K}(C) = q^{\mathrm{asc}(C)} \sum_{\theta\in C} \mathrm{e}_{\lambda(\theta)}(\mathbf{x})$$

to $\mathrm{LLT}_\Gamma(\mathbf{x}; q+1)$, which is elementary-positive. Summing over all classes we obtain the following.

**Corollary 2.12.** *For every circular decorated unit-interval graph $\Gamma$,*

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = \sum_{\theta\in\mathcal{O}(\Gamma)} q^{\mathrm{asc}(\theta)} \mathrm{e}_{\lambda(\theta)}(\mathbf{x}).$$

*In particular,* $\mathrm{LLT}_\Gamma(\mathbf{x}; q+1)$ *is elementary-positive.*

For ordinary graphs this is Theorem 2.4. Our proof does not use the recursion of [AS22], and thus gives a new proof of Theorem 2.4.

## 3. The four-term identity

This section contains the only local identity used in the proof, together with two facts about highest reachable vertices. We state them for sets of edges rather than orientations, since later we apply them to sets obtained by adding or deleting single edges.

Let $\eta$ be a set of pairs $uv$ with $u < v$ in $[n]$, regarded as edges directed upwards, or a translation-invariant set of such pairs of integers whose projection to $\mathbb{Z}/n\mathbb{Z}$ is acyclic. In both cases every vertex $z$ has a highest reachable vertex $\mathrm{hrv}(\eta, z)$; in the periodic case, the bound follows as in Section 2.5. The map $\mathrm{hrv}(\eta, \cdot)$ is weakly decreasing along every edge of $\eta$, and $\mathrm{hrv}(\eta, z)$ is the largest sink, that is, vertex without outgoing edges, which can be reached from $z$.

**Lemma 3.1.** *Let $\eta$ be as above.*

*(i) Let $\xi$ be a set of upward pairs, translation invariant in the periodic case, such that $\mathrm{hrv}(\eta, y) \le \mathrm{hrv}(\eta, x)$ for every $xy \in \xi$. Then $\mathrm{hrv}(\eta \cup \xi, \cdot) = \mathrm{hrv}(\eta, \cdot)$, and in the periodic case $\eta \cup \xi$ still has acyclic projection.*

*(ii) If $\eta$ contains a path from $x$ to $y$ avoiding the edge $xy$, then deleting $xy$ does not change reachability.*

*Proof.* For (i), $\mathrm{hrv}(\eta, \cdot)$ is weakly decreasing along every edge of $\eta \cup \xi$. Hence every vertex reachable from $z$ in $\eta \cup \xi$ is at most $\mathrm{hrv}(\eta, z)$, and the maximizing path in $\eta$ is still available. In the periodic case, a directed cycle in the projection would lift to a path from $z$ to $z + kn$ for some $k \geq 1$, and then $\mathrm{hrv}(\eta, z) + kn = \mathrm{hrv}(\eta, z + kn) \leq \mathrm{hrv}(\eta, z)$, which is absurd. For (ii), every path using $xy$ can be rerouted along the alternative path. ∎

**Lemma 3.2.** *Let $\eta$ and $\eta'$ be sets of upward pairs in $[n]$, and let $\sigma$ be a permutation of $[n]$ mapping $\eta$ bijectively onto $\eta'$. If $\sigma$ is increasing on the sinks of $\eta$, then $\mathrm{hrv}(\eta', \sigma(z)) = \sigma(\mathrm{hrv}(\eta, z))$ for all $z$. In particular the fibres of $\mathrm{hrv}(\eta', \cdot)$ are the images of the fibres of $\mathrm{hrv}(\eta, \cdot)$.*

*Proof.* The vertex $\mathrm{hrv}(\eta, z)$ is the largest sink of $\eta$ reachable from $z$, and $\sigma$ maps the sinks of $\eta$ reachable from $z$ onto the sinks of $\eta'$ reachable from $\sigma(z)$, preserving their order. ∎

**Lemma 3.3** (The four-term identity)**.** *Let $\eta$ be a set of upward pairs in $[n]$. Let $u < v < w$ be vertices such that $uw \in \eta$, and $uv, vw \notin \eta$. Put $\eta_{00} = \eta$, $\eta_{10} = \eta + uv$, $\eta_{01} = \eta + vw$ and $\eta_{11} = \eta + uv + vw$, and write $\mathrm{K}(\eta_{ij})$ for $\mathrm{K}$ of the transitive closure.*

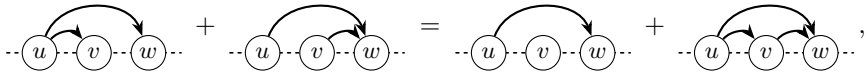
*(i) For every coloring $\kappa : [n] \to \mathbb{N}^+$ which increases along the edges of $\eta$,*

$$[\kappa \text{ increases on } \eta_{10}] + [\kappa \text{ increases on } \eta_{01}] = 1 + [\kappa \text{ increases on } \eta_{11}].$$

*In particular $\mathrm{K}(\eta_{10}) + \mathrm{K}(\eta_{01}) = \mathrm{K}(\eta_{00}) + \mathrm{K}(\eta_{11})$, and the same holds for the sums over colorings with prescribed colors at any set of vertices.*

*(ii) The four sets can be paired, either as $\{\eta_{00}, \eta_{01}\}$ and $\{\eta_{10}, \eta_{11}\}$ or as $\{\eta_{00}, \eta_{10}\}$ and $\{\eta_{01}, \eta_{11}\}$, so that paired sets have the same highest reachable vertices. In particular $\Phi(\eta_{10}) + \Phi(\eta_{01}) = \Phi(\eta_{00}) + \Phi(\eta_{11})$ for every function $\Phi$ of the highest reachable vertex map, for example $\Phi = \mathrm{e}_\lambda$ of its fibre sizes.*

The lemma has no cover hypotheses, and the other edges of $\eta$ are arbitrary. Writing each set for its value, the identity reads




where the dotted line stands for the other vertices and edges, which are the same in all four graphs.

*Proof.* For (i), write $a, b, c$ for the colors at $u, v, w$, so that $a < c$. Then $[a < b] + [b < c] = 1 + [a < b < c]$, which is the displayed identity. Summing $x_\kappa$ over all $\kappa$ increasing on $\eta$, with any colors prescribed, gives the second statement.

For (ii), write $h(z) = \mathrm{hrv}(\eta, z)$. Since $uw \in \eta$, $h(w) \leq h(u)$. Suppose first that $h(w) \leq h(v)$. Adding $uv$ does not change the highest reachable vertices of $v$ and $w$, since no path starting at $v$ or $w$ uses an edge leaving $u < v$. Hence $\mathrm{hrv}(\eta_{10}, w) \leq \mathrm{hrv}(\eta_{10}, v)$ as well, and Lemma 3.1(i) for $\xi = \{vw\}$ pairs $\eta_{00}$ with $\eta_{01}$ and $\eta_{10}$ with $\eta_{11}$. Suppose instead that $h(v) < h(w)$. Then $h(v) < h(w) \leq h(u)$. In $\eta_{01}$, a path from $v$ either avoids $vw$ or continues from $w$, so $\mathrm{hrv}(\eta_{01}, v) \leq \max(h(v), h(w)) = h(w) \leq \mathrm{hrv}(\eta_{01}, u)$. Lemma 3.1(i) for $\xi = \{uv\}$, applied to $\eta$ and to $\eta_{01}$, gives the second pairing. ∎

Lemma 3.3 does not say that the two middle terms correspond to each other. In the first case of the proof, $\eta_{10}$ has the same highest reachable vertices as $\eta_{11}$, not as $\eta_{01}$. This is why the identity is used for sums over many orientations at once.

## 4. Flip classes of ordinary graphs

In this section we prove Theorem 2.11 for ordinary graphs, assuming a connectivity statement, Theorem 4.9, which we prove in Section 5.

The idea is as follows. The four-term identity gives linear relations among the sums $\mathsf{K}(\mathcal{L}_s)$ and $\mathsf{E}(\mathcal{L}_s)$ over all orientations with $s$ strict and ascending edges. These are the bounce relations of [AS22] in a form that keeps the number of edges fixed. To pass from all orientations to one flip class $C$, we show that every other class with the same number of edges can be written, with the same integer coefficients for $\mathsf{K}$ and $\mathsf{E}$, in terms of classes of smaller graphs. For this we need to know that one class contains all orientations of a certain kind; this is Theorem 4.9. The relation is first obtained for an initial piece of the graph, called the prefix, and then transported to the whole graph by fixing the color of the last vertex of the prefix. Throughout, $\Gamma$ is an ordinary decorated unit-interval graph on $[n]$, and $|E(\Gamma)| = \sum_i a_i$ denotes its number of natural edges.

4.1. **Changing the decoration.** The values $\mathsf{K}$ and $\mathsf{E}$ only depend on the posets $P_+(\theta)$, and the flip relation only depends on the graphs $\theta \cup S$. Hence the following two operations transport flip classes to flip classes and preserve both values.

**Lemma 4.1.** *Let $C$ be a flip class of $\Gamma$ and let $f$ be a corner.*

*(i) If $f$ is an edge of $\theta \cup S$ for all $\theta \in C$, then marking $f$ strict, or unmarking it if it is strict, gives a graph $\Gamma'$ in which $C$ is again a flip class, with the same graphs $\theta \cup S$.*

*(ii) If $f$ is an edge of $\theta \cup S$ for no $\theta \in C$, then deleting $f$ gives a decorated unit-interval graph $\Gamma \setminus f$ with $|E(\Gamma)| - 1$ natural edges, in which $C$ is again a flip class.*

*Proof.* A corner is never a short edge of a flip. In (i) we therefore only change the label of an edge which is present throughout, and the flips are the same. In (ii), deleting the maximal interval $f$ keeps the natural edges closed under subintervals, and the remaining strict edges are still corners. No orientation in $C$ uses $f$, so $f$ is not the long edge of a flip from $C$, and conversely every flip of $\Gamma \setminus f$ is a flip of $\Gamma$. ∎

By Lemma 2.8, every corner is either present throughout a flip class or absent throughout.

4.2. **Fixing the color of a vertex.** We shall need the two values with the color of one vertex $c$ fixed. For an orientation $\theta$ and $\beta \in \mathbb{N}^+$, let $\mathrm{K}_\theta[\beta]$ be the sum of $x_\kappa$ over the strict $P_+(\theta)$-partitions $\kappa$ with $\kappa(c) = \beta$. For the elementary value, replace every fibre of $\mathrm{hrv}(\theta, \cdot)$ by the chain on its elements in increasing order. The strict partitions of this disjoint union of chains give $\mathrm{e}_{\lambda(\theta)}$, and we let $\mathrm{e}_\theta[\beta]$ be the sum over those with $\kappa(c) = \beta$. If $c$ is the largest vertex of its fibre, which has $p$ elements, then

$$\mathrm{e}_\theta[\beta] = x_\beta \, \mathrm{e}_{p-1}(x_1, \dots, x_{\beta-1}) \prod_B \mathrm{e}_{|B|}(\mathbf{x}), \tag{3}$$

the product ranging over the other fibres $B$. For a set $\Omega$ of orientations we write $\mathsf{K}(\Omega)[\beta]$ and $\mathsf{E}(\Omega)[\beta]$ for the corresponding sums. Summing over $\beta$ gives back $\mathsf{K}(\Omega)$ and $\mathsf{E}(\Omega)$.

Below, $c$ is always the largest vertex of the graph under consideration, so that (3) applies, and $\mathrm{e}_\theta[\beta]$ only depends on the fibre sizes and on the size of the fibre of $c$.

The values with the color of $c$ fixed are preserved by every operation used below: by Lemma 4.1; by deleting a redundant edge, which preserves $P_+(\theta)$; by the four-term identity, whose pairing preserves the whole map $\mathrm{hrv}(\theta, \cdot)$; and by relabellings as in Lemma 3.2 which fix $c$.

4.3. **Two relations.** For an integer $s$, let $\mathcal{L}_s(\Gamma)$ be the set of orientations $\theta$ such that $\theta \cup S$ has $s$ edges. By Lemma 2.8, $\mathcal{L}_s(\Gamma)$ is a union of flip classes. We use two local configurations, see Figure 4.

**Definition 4.2.** Let $x < y < t$ be vertices with $y = x + 1$, such that $xt$ is a strict edge, $y$ is not the endpoint of any strict edge, and $xt$ is the only strict edge leaving $x$. Assume that no natural edge joins a vertex at most $y$ to a vertex larger than $t$, that is, $b(t+1) > y$, where we put $b(n+1) = n+1$.

(i) The triple $(x, y, t)$ is an *NN contact* if $b(x) = b(y)$ and no strict edge ends at $x$. Its *target* $R$ is obtained by deleting the edge $xt$ and marking the corner $yt$ strict.

(ii) Let $u < x$. The quadruple $(u, x, y, t)$ is a *DN contact* if $ux$ is a strict edge, no other strict edge ends at $x$, and $b(y) = u + 1$. Its *target* $R$ is obtained by changing $a_y$ to $a_y + 1$ and $a_t$ to $a_t - 1$, so that $uy$ becomes a natural edge and $xt$ does not, and by replacing the strict edges $ux, xt$ by $uy, yt$.

One checks directly from the inequalities that $R$ is a decorated unit-interval graph. In the DN case, $a_y = x - u = a_x$, so the new area sequence is valid at $y$, and $a_{t+1} < a_t$ by the condition on $b(t+1)$. The target of an NN contact has one natural edge fewer. The target of a DN contact has the same number of natural edges, and $\mathrm{mom}(\Gamma) := \sum_i i\, a_i$ is smaller by $t - y$.

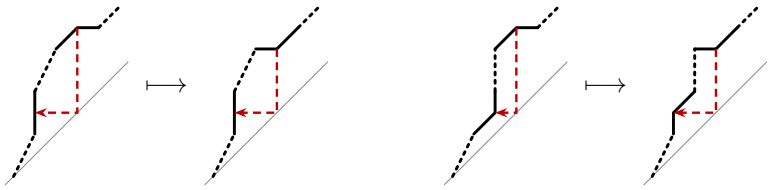


FIGURE 4. An NN contact (left) and a DN contact (right), each followed by its target, drawn as in [AS22, (3), (4)]. Thick segments are steps of the Schröder path, and dotted segments are arbitrary. The diagonal steps at the top cross the strict edge $xt$, and $yt$ in the targets. The dashed red path goes down from there to the diagonal box of $x$ and then left to the path, between the rows of $x$ and $y = x + 1$. In the DN contact the diagonal step at the bottom crosses the strict edge $ux$, and in its target $uy$.

**Proposition 4.3.** *Let $\Psi$ be one of $\mathsf{K}$, $\mathsf{E}$, $\mathsf{K}[\beta]$ or $\mathsf{E}[\beta]$, with the color fixed at $c = n$. If $\Gamma$ has an NN contact with target $R$, then for all $s$*

$$\Psi(\mathcal{L}_s(\Gamma)) = \Psi(\mathcal{L}_s(R)) + \Psi(\mathcal{L}_{s-1}(R)).$$

*If $\Gamma$ has a DN contact with target $R$, then $\Psi(\mathcal{L}_s(\Gamma)) = \Psi(\mathcal{L}_s(R))$ for all $s$.*

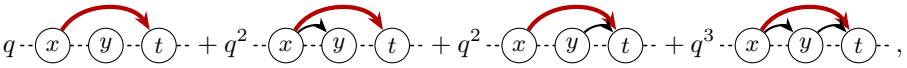
*Proof.* For an orientation $\theta$ we write $\Psi(\theta) = \Psi(\{\theta\})$. Comparing coefficients of $q^s$, the two relations are equivalent to

$$\sum_{\theta\in\mathcal{O}(\Gamma)} q^{|\theta\cup S|}\,\Psi(\theta) = (1+q)\sum_{\theta\in\mathcal{O}(R)} q^{|\theta\cup S|}\,\Psi(\theta) \tag{4}$$

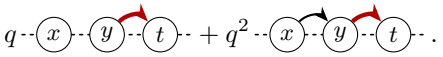
for an NN contact, and to the same identity without the factor $1+q$ for a DN contact; here $q$ is a formal variable.

Consider an NN contact $(x, y, t)$, and let $\sigma$ be the transposition of $x$ and $y$. Let $E'$ be the set of natural edges other than $xy$, $yt$ and $xt$. It is the same for $\Gamma$ and $R$, and $\sigma$ permutes it: for $p < x$, the pair $px$ is natural if and only if $py$ is, since $b(x) = b(y)$; for $y < p < t$, the pairs $xp$ and $yp$ are both natural since they lie inside $xt$; and there are no natural edges from $x$ or $y$ to vertices beyond $t$. The strict edges in $E'$ avoid $x$ and $y$, so $\sigma$ fixes them.

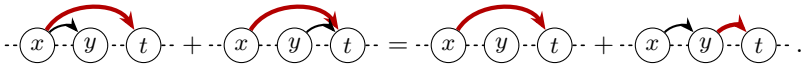
We group the orientations $\theta$ of $\Gamma$ and of $R$ by the set $\eta = (\theta \cup S) \cap E'$. In the pictures below, the vertices $x < y < t$ are drawn in increasing order, the dotted line stands for $\eta$, thick red arrows are strict edges and thin black arrows are ascending edges, and each picture stands for the value $\Psi$ of the corresponding set of edges. For fixed $\eta$, the orientations of $\Gamma$ contribute $q^{|\eta|}$ times



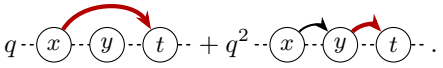

and the orientations of $R$ contribute $q^{|\eta|}$ times




By the four-term identity, and since $xt$ is redundant once $xy$ and $yt$ are present,



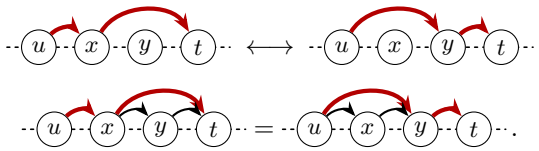

Hence the orientations of $\Gamma$ with fixed $\eta$ contribute $(1+q)\,q^{|\eta|}$ times




Finally, $\sigma$ maps $\eta + xt$ onto $\sigma(\eta) + yt$. Its only inversion is the pair $x < y$, and $x$ is not a sink of $\eta + xt$, so Lemma 3.2 applies; $\sigma$ fixes $c = n$, and relabelling colorings by $\sigma$ preserves K. Hence the first picture for $\eta$ has the same value as the first picture of $R$ for $\sigma(\eta)$. Since $\sigma$ permutes the possible sets $\eta$ and preserves their sizes, summing over $\eta$ gives (4).

For a DN contact $(u, x, y, t)$, let $E'$ be the set of natural edges of $\Gamma$ or $R$ other than $ux$, $uy$, $xy$, $xt$ and $yt$, and again let $\sigma$ exchange $x$ and $y$. The previous argument shows that $\sigma$ permutes $E'$, now using that for $p < x$ with $p \neq u$, the pair $px$ is natural if and only if $py$ is. With the vertices $u < x < y < t$, the source has the strict edges $ux, xt$ and the target the strict edges $uy, yt$, and the two remaining local edges of each may be ascending or not. We apply the four-term identity to the triangle $x < y < t$ in the source and to the triangle $u < x < y$ in the target. It

remains to compare the terms with no ascending local edge, and with two:




In the first line, $\sigma$ maps $\eta + ux + xt$ onto $\sigma(\eta) + uy + yt$, and $x$ is not a sink, so the values agree after replacing $\eta$ by $\sigma(\eta)$. In the second line, both sets contain the path $u \to x \to y \to t$, and their remaining local edges $xt$ and $uy$ are redundant. As in the NN case, the source and the target then contribute $(1+q)\,q^{|\eta|+2}$ times the same two values, after reindexing by $\sigma$, which proves the second relation. ∎

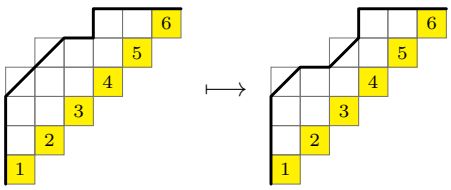
Multiplying (4) by $q^{-|S|}$, and using that $\Gamma$ and $R$ have the same number of strict edges, the NN relation becomes $F(\Gamma) = (1+q)F(R)$ and the DN relation becomes $F(\Gamma) = F(R)$, for $F(\Gamma) = \mathrm{LLT}_\Gamma(\mathbf{x}; q+1)$ and for the right-hand side of Theorem 2.4. These correspond to the shifted forms of the first two bounce relations of [AS22, Thm. 2.1] for a single bounce point.

**Remark 4.4.** For comparison, we collect the relations of [AS22, Thm. 2.1 and Thm. 2.5] after the substitution $q \mapsto q+1$. There $\Gamma$ is encoded by a Schröder path $P$ in the steps $\mathtt{n}, \mathtt{d}, \mathtt{e}$, and we write $F(P) = \mathrm{LLT}_P(\mathbf{x}; q+1)$. Then $F(\mathtt{nd}^k\mathtt{e}) = \mathrm{e}_{k+1}$, $F$ is multiplicative under concatenation, and

$$\begin{aligned}
F(U\mathtt{ne}V) &= F(U\mathtt{en}V) + qF(U\mathtt{d}V) && \text{(corner)},\\
F(U\mathtt{n.n}V\mathtt{d.e}W) &= (1+q)F(U\mathtt{n.n}V\mathtt{e.d}W) && \text{(NN)},\\
F(U\mathtt{d.n}V\mathtt{d.e}W) &= F(U\mathtt{n.d}V\mathtt{e.d}W) && \text{(DN)},\\
F(U\mathtt{n.d}V\mathtt{d.e}W) &= qF(U\mathtt{n.d}V\mathtt{e.d}W) + (1+q)F(U\mathtt{d.n}V\mathtt{e.d}W) && \text{(ND)},
\end{aligned}$$

where in the last three relations $V \in \{\mathtt{n}, \mathtt{d}\}^*$ and the bounce path at the marked points has a single bounce point. The corner relation deletes a normal corner, or marks it strict; it follows by summing Lemma 4.1 over all flip classes, since every corner is present throughout a flip class or absent from it. The NN and DN relations are the summed forms of Proposition 4.3. We do not use the ND relation: in Lemma 4.13, a strict corner ending at $y$ is unmarked instead, by Lemma 4.1(i).

**Example 4.5.** In the graph $\Gamma$ of Example 2.1, with path $\mathtt{nnnddeneee}$, area sequence $\mathbf{a} = (0,1,2,3,3,2)$ and strict edges 14 and 25, the triple $(2,3,5)$ is an NN contact: 25 is the only strict edge leaving 2, no strict edge ends at 2 or 3, $b(2) = b(3) = 1$, and $b(6) = 4 > 3$. In the notation of Remark 4.4, the path is $U\mathtt{n.n}V\mathtt{d.e}W$ with $U = \mathtt{n}$, $V = \mathtt{d}$ and $W = \mathtt{neee}$. Its target $R$ has the path $U\mathtt{n.n}V\mathtt{e.d}W = \mathtt{nnndedneee}$, with area sequence $(0,1,2,3,2,2)$ and strict edges 14 and 35:




The graph $\Gamma$ has nine normal edges and $R$ has eight, and Proposition 4.3, summed as above, gives $\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = (1+q)\,\mathrm{LLT}_R(\mathbf{x}; q+1)$.

4.4. **Three smaller classes.** We now express certain flip classes in terms of flip classes of graphs with fewer natural edges. In all three cases the coefficients are the same for $\mathsf{K}$ and $\mathsf{E}$, and for the values with the color fixed at $c = n$.

**Lemma 4.6.** *If some corner is absent from $\theta \cup S$ for $\theta \in C$, then $C$ is a flip class of a graph with fewer natural edges, with the same values.*

*Proof.* This is Lemma 4.1(ii). ∎

**Lemma 4.7.** *Suppose that the natural graph of $\Gamma$ is connected, but the underlying undirected graph of $\theta \cup S$ is disconnected for $\theta \in C$. Then there is a graph $\Gamma'$ with fewer natural edges on the same vertex set, and a flip class $C'$ of $\Gamma'$, with the same values, also with the color fixed at $c = n$.*

*Proof.* The connected components $B_1, \dots, B_r$ of $\theta \cup S$ are the same for all $\theta \in C$. List them with the component of $n$ last and the others by their least elements, and let $\rho$ be the permutation of $[n]$ placing them as consecutive intervals in this order, keeping the order inside each component. Let $\Gamma'$ be the disjoint union of the graphs induced by $\Gamma$ on the components, placed on these intervals. An induced subgraph of a unit-interval graph on a subset, with the inherited order, is again a unit-interval graph, and its strict edges are still corners. Since the natural graph of $\Gamma$ is connected, some natural edge joins two components, and it is lost; hence $|E(\Gamma')| < |E(\Gamma)|$.

Every edge of $\theta \cup S$, every flip, and every directed path between two vertices of a component lies inside that component. Hence $\rho$ maps $C$ bijectively onto a flip class $C'$ of $\Gamma'$, and the cover tests agree. Relabelling colorings by $\rho$ preserves K. The highest reachable vertex of $z$ lies in the component of $z$, and $\rho$ is increasing on each component; so $\rho$ maps fibres to fibres, and fixes $n$ together with the position of $n$ in its fibre. ∎

**Lemma 4.8.** *Suppose that every corner is strict, and that for every $\theta \in C$ some corner is not a cover of $P_+(\theta)$. For a set $T$ of corners, let $\Gamma \setminus T$ be the graph with the corners in $T$ deleted. Then the orientations in $C$, viewed as orientations of $\Gamma \setminus T$, form a union $\mathcal{P}_T(C)$ of flip classes, and*

$$\Psi(C) = \sum_{\emptyset \neq T} (-1)^{|T|+1} \sum_{C' \in \mathcal{P}_T(C)} \Psi(C')$$

*for each of the values $\Psi$ above, the outer sum ranging over nonempty sets of corners.*

*Proof.* Let $e$ be a natural edge and $f \neq e$ a corner. A directed path between the endpoints of $e$ stays inside the interval of $e$, and $f$ is not contained in that interval, since $e$ lies inside some corner and $f$ is maximal. So deleting corners other than $e$ does not change whether $e$ is a cover. The orientations of $\Gamma \setminus T$ are the same sets of ascending edges, and the flips of $\Gamma \setminus T$ are exactly the flips of $\Gamma$ whose long edge is not in $T$. Hence the image of $C$ is closed under flips of $\Gamma \setminus T$, that is, it is a union of flip classes.

Fix $\theta \in C$ and a corner $f$ which is not a cover of $P_+(\theta)$. The alternative path between the endpoints of $f$ lies strictly inside $f$ and uses no corner. It survives the deletion of any set of corners, so by Lemma 3.1(ii) the sets $T$ and $T \cup \{f\}$ give the same reachability whenever $f \notin T$. Pairing these sets gives $\sum_T (-1)^{|T|} \Psi((\theta \cup S) \setminus T) = 0$, the sum ranging over all sets of corners. The corner $f$ may depend on $\theta$, but the identity holds for each $\theta$. Summing over $C$ and separating the term $T = \emptyset$ proves the claim. ∎

4.5. **One flip class.** The following theorem is proved in Section 5. A vertex $v$ is an *articulation* of a connected graph if deleting it disconnects the graph.

**Theorem 4.9.** *Suppose that the natural graph of $\Gamma$ consists of a path $1, 2, \dots, b$ followed by a connected graph on $[b, n]$ without articulations, where the path may consist of the single vertex $1 = b$. Let $\theta, \theta'$ be orientations with the same number of ascending edges such that the underlying graphs of $\theta \cup S$ and $\theta' \cup S$ are connected and every corner is a cover of both $P_+(\theta)$ and $P_+(\theta')$. Then $\theta$ and $\theta'$ lie in the same flip class.*

We shall express flip classes in the form

$$\Psi(C) = \sum_i z_i V(C_i) \qquad \text{for } \Psi \in \{\mathsf{K}, \mathsf{E}, \mathsf{K}[\beta], \mathsf{E}[\beta]\}, \tag{5}$$

with the color fixed at $c = n$, where the integers $z_i$ do not depend on $\Psi$, and every $C_i$ is a flip class of a graph $\Gamma_i$ on $[n]$ such that $(|E(\Gamma_i)|, \mathrm{mom}(\Gamma_i))$ is lexicographically smaller than $(|E(\Gamma)|, \mathrm{mom}(\Gamma))$.

**Proposition 4.10.** *Suppose that $\Gamma$ satisfies the hypothesis of Theorem 4.9 and has an NN or a DN contact. Let $C$ be a flip class containing an orientation $\theta$ such that $\theta \cup S$ is connected and every corner is a cover of $P_+(\theta)$. Then $C$ has an expansion* (5).

*Proof.* Let $s$ be the number of edges of $\theta \cup S$. Every other flip class $C' \subseteq \mathcal{L}_s(\Gamma)$ omits a corner, or has a disconnected graph, or contains no orientation in which all corners are covers; otherwise it would contain an orientation as in Theorem 4.9, and hence equal $C$. In the first two cases Lemmas 4.6 and 4.7 give an expansion (5) of $C'$. In the third case, all corners are present in $C'$, and we mark them strict by Lemma 4.1(i) and apply Lemma 4.8. Proposition 4.3 expresses $\Psi(\mathcal{L}_s(\Gamma))$ in terms of one or two of the sets $\mathcal{L}_r(R)$ of the target $R$, which are unions of flip classes of $R$. Hence

$$\Psi(C) = \Psi(\mathcal{L}_s(\Gamma)) - \sum_{C' \neq C} \Psi(C')$$

is an expansion (5). ∎

4.6. **Gluing at one vertex.** Suppose that $c \in [n]$ is such that no natural edge joins a vertex smaller than $c$ to a vertex larger than $c$. Let $\Gamma_{\leq c}$ and $\Gamma_{\geq c}$ be the graphs induced on $[1, c]$ and on $[c, n]$; we say that $\Gamma$ is obtained by *gluing* them at $c$. An orientation of $\Gamma$ is a pair of orientations of the two parts. A directed path between two vertices of one part stays in that part, and a triangle $u < v < w$ with $uw$ natural lies in one part. Hence the flip classes of $\Gamma$ are the products $C_1 \times C_2$ of flip classes of the two parts.

**Lemma 4.11.** *Let $C_2$ be a flip class of $\Gamma_{\geq c}$, and suppose that, with the color fixed at $c$,*

$$\Psi(C_1)[\beta] = \sum_i z_i V(C_{1,i})[\beta] \qquad \textit{for all } \beta \textit{ and } \Psi \in \{\mathsf{K}, \mathsf{E}\},$$

*where $C_{1,i}$ is a flip class of a graph $\Gamma_{1,i}$ on $[1, c]$. Then $\Psi(C_1 \times C_2) = \sum_i z_i V(C_{1,i} \times C_2)$ for $\Psi \in \{\mathsf{K}, \mathsf{E}\}$, where $C_{1,i} \times C_2$ is a flip class of the graph obtained by gluing $\Gamma_{1,i}$ and $\Gamma_{\geq c}$ at $c$.*

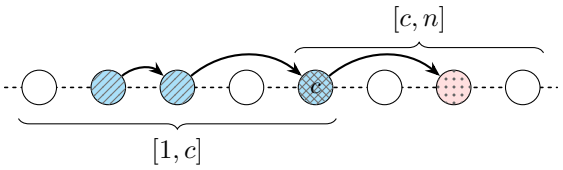



FIGURE 5. Gluing at $c$. A left fibre whose highest reachable vertex is $c$ (hatched) merges with the right fibre containing $c$ (dotted); $c$ lies in both. In the merged chain, $c$ is the largest left vertex and the smallest right vertex, and summing over its color $\beta$ gives $\sum_\beta x_\beta \, \mathrm{e}_{p-1}(x_1, \dots, x_{\beta-1}) \, \mathrm{e}_{r-1}(x_{\beta+1}, \dots) = \mathrm{e}_{p+r-1}$.

*Proof.* The glued graphs are decorated unit-interval graphs: a corner of the left part ends at most at $c$ and a corner of the right part starts at least at $c$, so neither contains the other. For orientations $\theta_1$ and $\theta_2$ of the two parts, a strict $P_+(\theta_1 \cup \theta_2)$-partition is a pair of strict partitions of the parts which agree at $c$. Hence

$$\mathrm{K}_{\theta_1 \cup \theta_2} = \sum_\beta \mathrm{K}_{\theta_1}[\beta] \, \widehat{\mathrm{K}}_{\theta_2}[\beta],$$

where $\widehat{\mathrm{K}}_{\theta_2}[\beta]$ is the sum over strict partitions of $P_+(\theta_2)$ with value $\beta$ at $c$, with the factor $x_\beta$ omitted.

A left vertex reaches the right part only through $c$. Hence the highest reachable vertex of a left vertex $u$ is $\mathrm{hrv}(\theta_1, u)$ if this is smaller than $c$, and $\mathrm{hrv}(\theta_2, c)$ otherwise, while the right vertices keep their highest reachable vertices. The fibre of $c$ in the left part, of size $p$, merges with the fibre of $c$ in the right part, of size $r$, and all other fibres are unchanged, see Figure 5. In the two chains, $c$ is the largest element of the first and the smallest element of the second. Since

$$\sum_\beta x_\beta \, \mathrm{e}_{p-1}(x_1, \dots, x_{\beta-1}) \, \mathrm{e}_{r-1}(x_{\beta+1}, x_{\beta+2}, \dots) = \mathrm{e}_{p+r-1}(\mathbf{x}),$$

we obtain $\mathrm{e}_{\lambda(\theta_1 \cup \theta_2)} = \sum_\beta \mathrm{e}_{\theta_1}[\beta] \, \widehat{\mathrm{e}}_{\theta_2}[\beta]$, with $\widehat{\mathrm{e}}_{\theta_2}[\beta]$ defined in the same way. Multiplying the given identity by $\widehat{\Psi}(C_2)[\beta]$ and summing over $\beta$ proves the lemma. ∎

If $\Gamma$ is glued at $c$ from parts with $e_1$ and $e_2$ natural edges, then $\Gamma$ has $e_1 + e_2$ natural edges, and $\mathrm{mom}(\Gamma) = \mathrm{mom}(\Gamma_{\le c}) + \mathrm{mom}(\Gamma_{\ge c}) + (c-1)e_2$, since the area sequence of the right part starts with $a_c = 0$. Hence replacing the left part by a graph which is smaller in $(|E|, \mathrm{mom})$ makes the glued graph smaller as well.

4.7. **The prefix.** Suppose that the natural graph of $\Gamma$ is connected, that is, $a_i > 0$ for $i > 1$. If no $a_i$ exceeds one, the natural graph is the path $1, 2, \dots, n$.

**Lemma 4.12.** *Suppose that the natural graph of $\Gamma$ is connected and not a path. Let $j$ be the least index with $a_j \ge 2$, put $b = j - 2$, and let $c \ge j$ be the least index such that $c = n$ or $a_{c+1} = 1$. Then $\Gamma$ is glued at $c$ from $\Gamma_{\le c}$ and $\Gamma_{\ge c}$, and the natural graph of $\Gamma_{\le c}$ is the path $1, \dots, b$ followed by a graph on $[b, c]$ without articulations.*

*Proof.* A vertex $v$ with $1 < v < n$ is an articulation of a connected unit-interval graph if and only if $a_{v+1} = 1$. Indeed, if $a_{v+1} = 1$ then $b(v+1) = v$, so no natural edge passes over $v$. If $a_{v+1} \ge 2$, the natural edge $(v-1)(v+1)$ bypasses $v$ in the path of edges $i(i+1)$, which are natural since all $a_i$ are positive. By the choice of

$j$ we have $a_i = 1$ for $1 < i < j$ and $a_j = 2$, so $b = b(j)$ and the vertices $2, \dots, b$ are articulations. For $i > b$ we have $a_i \leq i - b$, so no natural edge ends above $b$ and starts below $b$. By the choice of $c$, no vertex strictly between $b$ and $c$ is an articulation of the graph induced on $[b, c]$, and $b(c+1) = c$ if $c < n$. ■

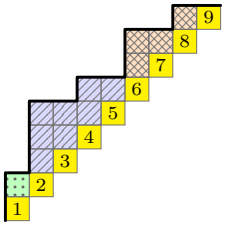



FIGURE 6. The area sequence $\mathbf{a} = (0, 1, 1, 2, 3, 2, 1, 2, 1)$, with $j = 4$, $b = 2$ and $c = 6$ in Lemma 4.12. The prefix is the path $1, 2$ (dotted) followed by the graph on $[2, 6]$ without articulations (hatched), and the whole graph is glued at 6 from the prefix and the graph on $[6, 9]$ (cross-hatched).

We call $\Gamma_{\leq c}$ the *prefix* of $\Gamma$, see Figure 6. For the prefix we put $a_{c+1} = 0$. The prefix has the same area sequence as $\Gamma$ up to $c$, and $a_{c+1} < a_c$ both in $\Gamma$ and in the prefix, since $a_c \geq 2$.

**Lemma 4.13.** *Let $\Gamma$ be a graph in which every corner is strict, with connected natural graph which is not a path. Let $t$ be the least index with $a_{t+1} < a_t$, where $a_{n+1} = 0$, and put $x = b(t)$ and $y = x + 1$. Then $a_t \geq 2$, and after unmarking the strict edge ending at $y$, if there is one, $(x, y, t)$ is an NN contact or $(b(x), x, y, t)$ is a DN contact.*

*Proof.* The areas $a_1, \dots, a_t$ are weakly increasing. If $a_t = 1$, then all these areas are at most one, and since $a_{t+1} = 0$, the natural graph would be a path; hence $a_t \geq 2$ and $x < y < t$. Since $a_{t+1} < a_t$, the edge $xt$ is a corner, hence strict, and $b(t+1) \geq x + 2 > y$.

For $i < t$ we have $b(i+1) - b(i) \in \{0, 1\}$, and a corner ends at $i$ exactly when $b(i+1) = b(i) + 1$. Hence the corners with heads at most $t$ have the tails $1, 2, \dots, x$, each exactly once, and the corners with heads above $t$ have tails at least $b(t+1) \geq x + 2$. In particular, no strict edge leaves $y$, and $xt$ is the only strict edge leaving $x$.

Since $y < t$, we have $a_x \leq a_y \leq a_x + 1$. If $a_y = a_x + 1$, then $b(x) = b(y)$, no corner ends at $x$, and $(x, y, t)$ is an NN contact once the corner ending at $y$, if any, is unmarked. If $a_y = a_x$, then $b(y) = b(x) + 1$. The pair $xy$ is natural, so $a_x = a_y \geq 1$, and $u = b(x)$ is the tail of the strict corner $ux$. Then $(u, x, y, t)$ is a DN contact once the corner ending at $y$, if any, is unmarked. ■

In the terminology of [AS22], the four cases are the contacts NN, ND, DN and DD at the first bounce point. By Lemma 4.1(i), unmarking the corner at $y$ in the ND and DD cases does not change a flip class in which this corner is present.

4.8. **Proof of Theorem 2.11 for ordinary graphs.** We proceed by induction on $n$, then on $|E(\Gamma)|$, then on $\mathrm{mom}(\Gamma)$. For $n \le 1$ there is only one orientation, and both values are 1 or $\mathrm{e}_1$.

Let $C$ be a flip class of $\Gamma$. If some corner is absent from $C$, we apply Lemma 4.6 and induction. Otherwise we mark all corners strict, by Lemma 4.1(i). If the natural graph is disconnected, it is a disjoint union of graphs on consecutive intervals; the flip class is a product of flip classes of these graphs, both values are products, and we use induction on $n$. If the natural graph is a path, all its edges are strict corners, there is only one orientation, and both values equal $\mathrm{e}_n$. If $\theta \cup S$ is disconnected for $\theta \in C$, or no orientation in $C$ has all corners as covers, we apply Lemma 4.7 or 4.8 and induction.

In the remaining case, let $\Gamma$ be glued at $c$ from its prefix and $\Gamma_{\ge c}$ as in Lemma 4.12, and write $C = C_1 \times C_2$. Let $\theta \in C$ be connected with all corners as covers. Its restriction to the prefix is connected, and every corner of the prefix is a corner of $\Gamma$ ending at most at $c$, whose cover test only involves vertices up to $c$. By Lemma 4.13 and Lemma 4.1(i), the prefix has an NN or DN contact, possibly after unmarking one corner which is present in $C_1$. The prefix satisfies the hypothesis of Theorem 4.9, and Proposition 4.10 gives an expansion (5) $\Psi(C_1)[\beta] = \sum_i z_i V(C_{1,i})[\beta]$ on $[1, c]$, with the color fixed at the last vertex $c$ of the prefix. By Lemma 4.11,

$$\Psi(C) = \sum_i z_i V(C_{1,i} \times C_2), \qquad \Psi \in \{\mathsf{K}, \mathsf{E}\}.$$

Each $C_{1,i} \times C_2$ is a flip class of a graph on $[n]$ which is smaller in $(|E|, \mathrm{mom})$, by the remark after Lemma 4.11. By induction $\mathsf{K}(C_{1,i} \times C_2) = \mathsf{E}(C_{1,i} \times C_2)$ for all $i$, and hence $\mathsf{K}(C) = \mathsf{E}(C)$. This completes the proof of Theorem 2.11 for ordinary graphs, given Theorem 4.9.

## 5. Connecting orientations by flips

In this section we prove Theorem 4.9. We first move the edges of an orientation into a normal form determined by the corners, Section 5.1. Section 5.2 constructs explicit paths of flips which exchange two edges and restore all others. With these we treat complete graphs and graphs with at most two corner chains, and then reduce the general case to smaller graphs by removing part of the last clique or the first vertex.

Throughout, $\Gamma$ is an ordinary decorated unit-interval graph, and we only consider orientations in which every corner is a cover. In particular every corner is an edge of $\theta \cup S$, and by Lemma 4.1(i) we may assume that all corners are strict. We write $uv \in \theta$ if $uv$ is an edge of $\theta \cup S$, and covers always refer to $P_+(\theta)$. We write

$$vy \mapsto yh \quad [vh]$$

for the flip which replaces the edge $vy$ by $yh$ and retains the long edge $vh$, and similarly when the short edges are exchanged in the other direction. By Lemma 2.6, such an exchange is a flip as soon as the two edges $vy$ and $vh$ are covers before it, and then $vh$ and $yh$ are covers after it. We repeatedly use the following observation: a directed path in $\theta \cup S$ between two vertices of an interval stays in that interval, so the cover test of an edge $uv$ only involves edges inside $[u, v]$.

5.1. **Corner chains.** Distinct corners have distinct tails and distinct heads, so the corners form disjoint directed paths, the *corner chains*; a vertex on no corner is a chain by itself. Let $r(u)$ be the last vertex of the corner chain containing $u$. A natural edge $uv$ with $u$ and $v$ on the same chain is a corner: if $uh$ is the corner leaving $u$, then $v \le h$ by maximality of $uh$, and $v \ge h$ since $v$ comes after $u$ on the chain. Let $\Gamma^+$ be the set of corners together with the normal edges $uv$ with $r(u) < r(v)$.

If $\theta \subseteq \Gamma^+$, that is, every edge of $\theta \cup S$ lies in $\Gamma^+$, then every corner is a cover. Indeed, $r$ is constant along corners and increases along the other edges of $\Gamma^+$, so a path between the endpoints $u, h$ of a corner only uses corners, and the only corner leaving $u$ is $uh$.

**Lemma 5.1.** *Every orientation $\theta$ in which all corners are covers is joined by flips to an orientation $\theta' \subseteq \Gamma^+$ which contains every edge of $\theta \cap \Gamma^+$.*

*Proof.* We process the corner chains in decreasing order of their last vertices. Let $U$ be the union of the chains already processed. We maintain that no edge of $\theta$ leaves $U$, and that all corners are covers when we start a new chain; initially $U = \emptyset$.

Let $B$ be the next chain, with last vertex $t$. Consider the edges $vy \in \theta$ with $v \in B$ and $r(y) < t$. For such an edge, $v < y \le r(y) < t$, so $v$ has a corner successor $h$ on $B$, and $y < h$ since $vy$ does not contain the corner $vh$. Choose a vertex $v$ with such an edge, and among its edges of this kind the edge $vy$ with the least head. Then $vy$ is a cover. Indeed, a path from $v$ to $y$ cannot pass through $U$, since no edge leaves $U$. An edge from $v$ to a vertex $z \notin U$ has $r(z) \le t$, by the choice of $t$. If $r(z) = t$, it is the corner $vh$, and $h > y$; if $r(z) < t$, it is one of the edges considered, and $z \ge y$ by the choice of $vy$. So the only path from $v$ to $y$ is the edge $vy$. Since $vh$ is a cover as well, we can perform the flip

$$vy \mapsto yh \quad [vh],$$

after which $vh$ is still a cover. The new edge $yh$ lies inside the corner $vh$, so it is normal, and $r(y) < r(h) = t$, so it lies in $\Gamma^+$ and is not one of the edges considered. The intervals of the other corners of $B$ have interiors disjoint from that of $vh$, so their cover tests are unchanged, and the edges inside $U$ are unchanged. Each flip removes one of the edges considered and creates none, so after finitely many flips there are none left.

Then no edge leaves $U \cup B$, and every edge created while processing $B$ ends in $B$. A path bypassing a corner outside $U \cup B$ cannot use such an edge, since it could not leave $U \cup B$ again; hence such a bypass would have existed before we processed $B$. Thus all corners are covers again. When all chains have been processed, all edges of the orientation lie in $\Gamma^+$, and only edges outside $\Gamma^+$ were removed. ∎

5.2. **Exchanging two edges.** The next lemma exchanges two edges along two crossing covers, restoring all other edges.

**Lemma 5.2.** *Let $a < b < c < d$, and let $\theta$ be an orientation in which $ac$ and $bd$ are covers, $ab \in \theta$ and $cd \notin \theta$. Then $\theta$ is joined by flips to the orientation $\theta - ab + cd$, in which $ac$ and $bd$ are again covers.*

*Proof.* All edges below lie inside $ac$ or $bd$, so they are natural and normal. Let $Y_a$ be the set of $v$ with $a < v < b$ and $av \in \theta$, and let $Y_d$ be the set of $w$ with $c < w < d$ and $wd \in \theta$. We perform the flips of Figure 7 in four stages:

(1) $av \mapsto vc$ $[ac]$ for $v \in Y_a$, in increasing order;
(2) $wd \mapsto bw$ $[bd]$ for $w \in Y_d$, in decreasing order;
(3) $ab \mapsto bc$ $[ac]$ and then $bc \mapsto cd$ $[bd]$;
(4) the flips of (2) in reverse, and then those of (1) in reverse.

By Lemma 2.6, each step is a flip as soon as its two edges are covers before it, and then the retained cover stays a cover. In (1), when $av$ is moved the edges from $a$ to smaller vertices have already been moved, so $av$ is a cover; these flips only change edges with tails below $b$, so $bd$ stays a cover. Stage (2) is symmetric, and only changes edges with heads above $c$. In (3), $ab$ is a cover since $a$ has no edge into $(a, b)$, and $bc \notin \theta$ since otherwise $ab, bc$ would bypass $ac$; after the first flip, $bd$ is still a cover, since a bypass would begin with $bc$ and no path leads from $c$ to $d$. In (4), the net change from $ab$ to $cd$ does not affect the edges inside $[b, w]$ and $[v, c]$, so every edge is a cover when it is moved back, as after the corresponding flip in (2) or (1). ■

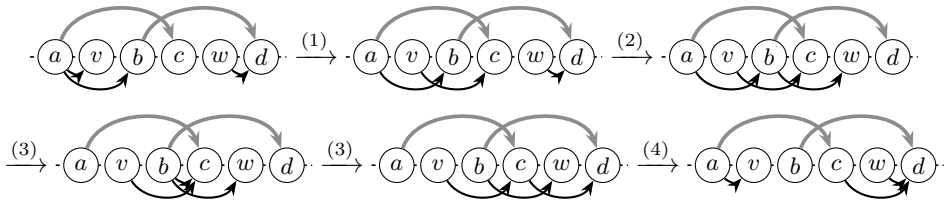



FIGURE 7. The flips in the proof of Lemma 5.2, with one vertex $v \in Y_a$ and one vertex $w \in Y_d$. The gray arcs above are the covers $ac$ and $bd$, which are retained throughout, the arcs below are the edges which move, and the dotted line stands for all other edges, which are restored at the end.

We now use Lemma 5.2 to move edges between two corner chains. Let $uv$ be a normal natural edge such that $u$ has a corner successor $h$. Then $v < h$, and $vh$ is natural since it lies inside $uh$. We call $vh$ the successor of the edge $uv$. Passing to the successor exchanges the two values of $r$ at the endpoints, and both endpoints increase. Every normal edge has at most one successor and at most one predecessor, so the normal edges form disjoint sequences in which each edge is followed by its successor.

**Lemma 5.3.** *For two corner chains there is at most one such sequence whose edges join the two chains. Along this sequence, the edges in $\Gamma^+$ and the edges not in $\Gamma^+$ alternate, and two consecutive edges in $\Gamma^+$ have the form $ab$ and $cd$ with $a < b < c < d$ such that $ac$ and $bd$ are corners.*

*Proof.* Let $uv$ be the first edge of such a sequence. If $v$ had a corner predecessor $p$, then $p < u$, since $uv$ does not contain the corner $pv$, and then $pu$ would be a predecessor of $uv$. Hence $v$ is the first vertex of its chain, and $u$ lies on a chain starting earlier. There is at most one such $u$ on a given chain: if $u < u' < v$ were two candidates, the natural edge $uv$ would contain the corner leaving $u$. This proves the first claim. Passing to the successor exchanges the two values of $r$, so membership in $\Gamma^+$ alternates. If $ab$ is followed by two further edges, they use the

corners leaving $a$ and $b$, say $ac$ and $bd$, and the path continues as $ab \to bc \to cd$ with $a < b < c < d$. ■

**Lemma 5.4.** *Let $\theta, \theta' \subseteq \Gamma^+$ be orientations which have, for every pair of corner chains, the same number of normal edges joining them. Then $\theta$ and $\theta'$ lie in the same flip class.*

*Proof.* List the positions in $\Gamma^+$ joining a given pair of chains in the order of their sequence. By Lemmas 5.3 and 5.2, we may replace an occupied position followed by a vacant one by the reverse, restoring all other edges. The sum of the indices of the occupied positions increases, so we reach the configuration occupying the last positions of the list, which only depends on the number of occupied positions. We do this for every pair of chains, starting from $\theta$ and from $\theta'$. ■

**Lemma 5.5.** *If there are at most two corner chains, then all orientations in which every corner is a cover and which have the same number of ascending edges lie in one flip class.*

*Proof.* By Lemma 5.1 we may assume that both orientations lie in $\Gamma^+$. With one chain, the only such orientation consists of the corners. With two chains, every normal edge of $\Gamma^+$ joins the two chains, and Lemma 5.4 applies. ■

5.3. **Complete graphs.** In this subsection we label the vertices $0, 1, \ldots, n$, with $n \geq 1$, and consider the complete natural graph. Its only corner is $0n$.

**Lemma 5.6.** *For the complete graph on $0, 1, \ldots, n$, the connected orientations in which $0n$ is a cover and which have the same number of ascending edges lie in one flip class.*

*Proof.* Let $P_n$ consist of $0n$ and the edges $i(i+1)$ for $1 \leq i < n$. We call the other pairs $ij$, with $1 \leq i < i+1 < j \leq n$, the chords of $P_n$. We first join every orientation in question to an orientation $P_n \cup X$, where $X$ is a set of chords, and then show that the orientations $P_n \cup X$ with the same number of chords lie in one flip class.

**Step 1: reaching $P_n \cup X$.** Here $r(0) = r(n) = n$ and $r(v) = v$ otherwise, so $\Gamma^+$ consists of $0n$ and all edges on $1, \ldots, n$. By Lemma 5.1 we may assume that $0n$ is the only edge at 0. Since $\theta$ is connected, the set $\eta$ of the remaining edges is connected on $1, \ldots, n$. Whenever a vertex $i$ has two outgoing covers $ij$ and $ik$ in $\eta$, with $j < k$, we perform the flip $ij \mapsto jk$ $[ik]$. The sum of the tails of the edges increases, so this terminates. Then every vertex has at most one outgoing cover. The covers form a connected graph, and a cycle in it would have a least vertex with two outgoing covers, so they form a tree, with unique sink $n$. Every edge of $\eta$ joins a vertex to one of its ancestors in this tree.

If $i < j$ are two children of $n$, their only outgoing edges are $in$ and $jn$, and the four flips

$$in \mapsto 0i \ [0n], \qquad jn \mapsto 0j \ [0n], \qquad 0i \mapsto ij \ [0j], \qquad 0j \mapsto jn \ [0n] \tag{6}$$

replace $in$ by $ij$, see Figure 8. For the first two, the vertex losing its edge to $n$ becomes a sink, so the new edge from 0 is a cover and $0n$ remains one. At the third flip, $i$ and $j$ are sinks; afterwards 0 is adjacent only to the sink $j$ and to $n$, which gives the last flip. In the tree, the branch $i \to n$ becomes $i \to j \to n$, and every edge still joins a vertex to an ancestor. The vertex $n-1$ is always a child of $n$. Applying (6) with $j = n-1$ until $n-1$ is the only child of $n$, the edges on

$1, \dots, n-1$ form a connected graph. Then we perform $(n-1)n \mapsto 0(n-1)$ $[0n]$ and apply the construction inductively on $0, \dots, n-1$. These flips remain flips in the whole graph, since a path between two vertices below $n$ cannot pass through $n$. At the end, 0 is adjacent only to $n-1$ and $n$, and $n-1$ has no outgoing edge, so we can perform $0(n-1) \mapsto (n-1)n$ $[0n]$. We have reached an orientation $P_n \cup X$.

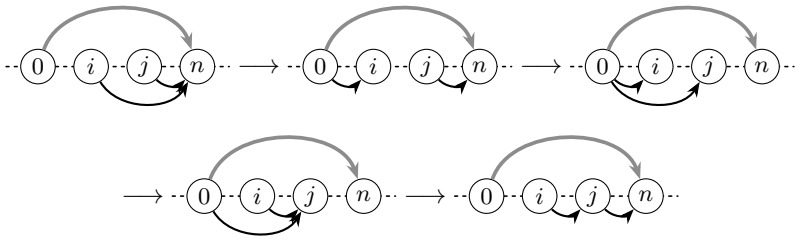



FIGURE 8. The four flips (6), which replace $in$ by $ij$; the cover $0n$ (gray) is retained throughout. The children $i$ and $j$ of $n$ become the chain $i \to j \to n$.

**Step 2: working in a smaller interval.** Let $0 < p \leq m \leq n$, and consider two orientations $P_n \cup X$ and $P_n \cup X'$ which agree on all pairs not contained in $W = \{0, p, p+1, \dots, m\}$. We show that a path of flips between their restrictions to $W$, in the complete graph on $W$, lifts to a path between them. First we open the chain by the flips

$$j(j+1) \mapsto 0j \quad [0(j+1)], \qquad j = n-1, n-2, \dots, m. \tag{7}$$

Before such a flip, all edges from 0 end at $j+1$ or above, so $0(j+1)$ is a cover, and these are flips regardless of the chords. Now the edges at 0 are $0m, 0(m+1), \dots, 0n$, and the restriction to $W$ is of the same form for the complete graph on $W$. A path between two vertices of $W$ stays in $W$: it cannot use a vertex below $p$, since no edge from 0 goes there, and it cannot return from above $m$. Hence flips in the complete graph on $W$ are flips of the whole graph. Opening $P_n \cup X'$ in the same way gives its restriction to $W$, with the same edges outside $W$, so after the path inside $W$ we close the chain by reversing the opening of $P_n \cup X'$.

**Step 3: rearranging the chords.** We show by induction on $n$ that the orientations $P_n \cup X$ with the same number of chords lie in one flip class. For $n \leq 3$ there is at most one chord. For $n = 4$ the chords are 13, 14 and 24. In the following table, a triple $ijk$ denotes the flip exchanging $ij$ and $jk$ across $ik$, in the direction allowed by the current state, and each row starts and ends with $P_4$ together with the indicated chords:

| Chords | Flips |
|---|---|
| $\{13\} \to \{14\}$ | $034, 023, 013, 014, 023, 034$ |
| $\{14\} \to \{24\}$ | $034, 124, 014, 023, 012, 023, 034$ |
| $\{13, 24\} \to \{14, 24\}$ | $034, 024, 012, 014, 024, 013, 023, 012, 023, 034$ |
| $\{14, 24\} \to \{13, 14\}$ | $034, 023, 012, 013, 023, 124, 034$ |

The cover conditions are checked directly at each step. These rows connect all sets of one or two chords.

Let $n \geq 5$. By Step 2 and induction applied to $W = \{0, 1, \ldots, n-1\}$ and to $W = \{0, 2, 3, \ldots, n\}$, we can rearrange the chords inside either set at a fixed number, restoring all other edges. The two sets of chord positions cover all positions except $1n$, and both contain 24. Hence we can exchange any two chords other than $1n$: if no single set contains both, we route through 24, moving the selected chord to 24 first if 24 is vacant, and moving 24 to the vacant position first otherwise. This connects all orientations $P_n \cup X$ with the same number of chords and the same status of $1n$.

It remains to change the status of $1n$. Suppose $1n \notin \theta$ and $13, 2n \in \theta$. We perform $in \mapsto 0i$ $[0n]$ for all $in \in \theta$ with $0 < i < n$, in decreasing order of $i$. When $in$ is moved, the edges into $n$ with larger tails have been moved. Hence $in$ is a cover, and the vertices other than $n$ adjacent to 0, which are larger than $i$, cannot reach $n$, so $0n$ is a cover. Now $02 \in \theta$ and $01 \notin \theta$, so 02 is a cover, and we perform the flip $12 \mapsto 01$ [02]. Then we reverse these flips in increasing order of $i$, including the new edge 01. When $0i$ is moved back to $in$, it has the least head among the edges at 0, and the edges into $n$ restored so far have smaller tails, so $0i$ and $0n$ are covers. We obtain $\theta - 12 + 1n$. To restore 12, we open the chain down to 3 by (7). On $0, 1, 2, 3$ the edges are then 03, 13 and 23, and (6) with $(i, j, n) = (1, 2, 3)$ replaces 13 by 12. Closing the chain gives

$$\theta \longmapsto \theta - 13 + 1n \qquad (13, 2n \in \theta,\ 1n \notin \theta). \tag{8}$$

Let $\theta = P_n \cup X$; there are $\binom{n-1}{2}$ chords in total. If $2 \leq |X| \leq \binom{n-1}{2} - 1$, we can first arrange $13, 2n \in \theta$ and $1n \notin \theta$, and (8) connects the two statuses of $1n$. If $|X| = 0$ or $|X| = \binom{n-1}{2}$, there is only one set of chords. If $|X| = 1$, start from $P_n + 13$, open the chain down to 2, perform $13 \mapsto 01$ [03] and $01 \mapsto 1n$ $[0n]$, and close the chain. While the chain is open, the vertices $2, \ldots, n$ are sinks, so these are flips, and we reach $P_n + 1n$. This completes the induction, and the proof. ∎

5.4. **Deleting a chain or an edge.** In this subsection, connectivity means connectivity of undirected graphs. Suppose that the natural graph is connected, has at least three vertices, and has no articulation. Then every pair $uv$ with $v - u \leq 2$ is a natural edge. Indeed, if $u(u+1)$ were not natural, then by closure under subintervals no natural edge would join $[1, u]$ and $[u+1, n]$; and if $u(u+2)$ were not natural, no natural edge would pass over $u+1$, which would be an articulation.

Replacing the natural graph by $\Gamma^+$ preserves connectivity within any union of corner chains. Indeed, if $uv$ is normal with $r(u) > r(v)$, and $uh$ is the corner leaving $u$, then $u < v < h$, the edge $vh$ lies in $\Gamma^+$, and the walk $u, h, v$ only visits the chains of $u$ and $v$.

**Lemma 5.7.** *Deleting the vertices of one corner chain leaves the other vertices connected in* $\Gamma^+$.

*Proof.* Two consecutive vertices $v, v+1$ are not on the same chain, since their edge would be a corner, while it lies inside $(v-1)(v+1)$ or $v(v+2)$. Hence the remaining vertices, listed in increasing order, have differences at most two, and consecutive ones are joined by natural edges. We replace the edges outside $\Gamma^+$ by the walks above, which avoid the deleted chain. ∎

**Lemma 5.8.** *If there are at least three corner chains, then deleting any normal edge $uv$ of $\Gamma^+$ leaves $\Gamma^+$ connected.*

*Proof.* Choose a vertex $w$ on a third chain. By Lemma 5.7, there is a path from $u$ to $w$ avoiding the chain of $v$, and a path from $w$ to $v$ avoiding the chain of $u$. Neither uses $uv$. ■

5.5. **The last clique.** Suppose that the last two corners are $bc$ and $an$, with $b < a < c < n$. Let $L = [1, c] \cup \{n\}$, and let $\Gamma[L]$ be the graph induced on $L$, with the inherited order and strict edges. For an orientation $\theta$ we write $\theta[L]$ for its edges inside $L$. All corners other than $an$ end at most at $c$, so the vertices in $(c, n)$ lie on no corner. Hence $\Gamma[L]$ has the same corners as $\Gamma$, its corner chains are those of $\Gamma$ except for the singleton chains of the vertices in $(c, n)$, and its set $\Gamma^+$ consists of the edges of $\Gamma^+$ inside $L$. Every vertex strictly between $a$ and $n$ is the last vertex of its chain. Hence an orientation in $\Gamma^+$ has no edge from $a$ other than $an$, and every natural edge with both endpoints strictly between $a$ and $n$, or ending at $n$ and starting after $a$, lies in $\Gamma^+$.

**Lemma 5.9.** *If Theorem 4.9 holds for $\Gamma[L]$, then it holds for $\Gamma$.*

*Proof.* Let $\theta$ be a connected orientation in which all corners are covers, and assume $\theta \subseteq \Gamma^+$ by Lemma 5.1. We show in four steps that $\theta$ can be joined to an orientation containing a fixed spanning tree $T$, and that all such orientations of the same size are joined.

**Components inside the last clique.** All pairs in $[a, n]$ are natural. Consider the connected components of the graph formed by the edges of $\theta$ inside $[a, n]$. A path between two vertices of such a component stays in $[a, n]$, and hence in the component. So a flip of the complete graph on a component, with its vertices in increasing order, is a flip of $\Gamma$ which changes the edges inside the component and fixes all other edges. The component containing $a$ and $n$ has the extreme edge $an$, which is a cover, so Lemma 5.6 applies to it.

**Step 1: making $\theta[L]$ connected.** In the component containing $a$ and $n$, the vertex $a$ is only adjacent to $n$. By Lemma 5.6 we replace the edges of this component by the star of all edges to $n$, together with other edges avoiding $a$ to keep their number. This is possible since a connected graph on $k$ vertices in which one vertex is a leaf has between $k - 1$ and $1 + \binom{k-1}{2}$ edges. Every other component meets $(a, c]$, since otherwise it would be a connected component of $\theta$. For such a component, let $u$ be its least vertex and $v \leq c$ another vertex. Take a path of covers from $u$ to $v$, using the edges in either direction. If its largest vertex $z$ exceeds $v$, its neighbours $i < j$ on the path are joined to $z$ by covers, and $jz \mapsto ij$ $[iz]$ is a flip, which shortens the path. Replacing an edge of the new path which is no longer a cover by a path of covers inside its interval, the largest vertex decreases. Repeating, we obtain a path of covers in $[u, v]$, which joins $u$ and $v$ by edges inside $L$. These flips do not disconnect pairs which were joined inside $[1, c]$ before. Repeating this, every component meets $L$ in a connected graph. Every natural edge with an endpoint in $(c, n)$ lies inside $an$, so we can replace every part of a path in $\theta$ outside $L$ by a path inside the corresponding component, and the edges of $\theta$ inside $L$ form a connected graph.

**Step 2: changing $\theta[L]$.** Suppose the edges of $\theta$ inside $L$ form a connected graph. We perform $vn \mapsto av$ $[an]$ for all $vn \in \theta$ with $c < v < n$, in decreasing order of $v$.

These are flips by the argument for the first flips in the proof of Lemma 5.2, and they do not change edges inside $L$ or the cover tests of other corners. Afterwards no vertex outside $L$ has an edge to $n$, so a path leaving $L$ cannot return to $L$. Hence every path of flips in $\Gamma[L]$ lifts, with all edges outside $L$ fixed. By assumption we can reach any connected orientation of $\Gamma[L]$ in $\Gamma^+$ with the same number of edges. At its end $a$ has no edge to $(a, n)$ inside $L$, and we reverse the first flips in increasing order of $v$.

**Step 3: a fixed tree.** In the set $\Gamma^+$ of $\Gamma[L]$, the corners and the edges $vn$ with $a < v \le c$ form a forest: every vertex has at most one outgoing edge among them, and all edges point upwards. Since $\theta[L]$ is connected, this forest extends to a spanning tree $T'$ of $\Gamma[L]$ inside $\Gamma^+$. Inside $[a, n]$, the tree $T'$ consists of the edges to $n$, and $T'$ does not contain $ba$, since $bc, cn, na$ already join $b$ and $a$. Adding edges of $\Gamma^+$ to $T'$ we obtain a connected orientation of $\Gamma[L]$ with the same number of edges as $\theta[L]$, and Step 2 reaches it. Now all of $[a, n]$ is one component. Indeed, every vertex in $(c, n)$ is joined, inside its component, to a vertex of $L$ by a path whose edges are not inside $L$, and these edges were not changed; and the vertices of $L \cap [a, n]$ are joined to $n$ by $T'$. Since $an$ is a cover, Lemma 5.6 makes this component a star at $n$ together with edges avoiding $a$, containing $T' \cap [a, n]$. This gives an orientation containing the spanning tree $T = T' \cup \{vn : c < v < n\}$, see Figure 9.

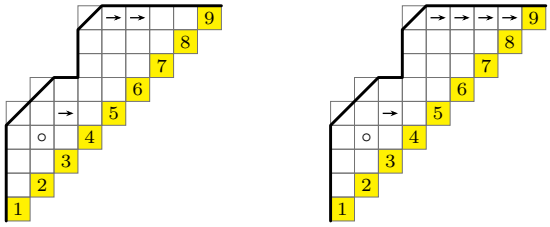



FIGURE 9. The trees $T'$ (left) and $T$ (right) for the path `nnnnddenndeeeee`, with corners 15, 26 and 49, so that $b = 2$, $a = 4$, $c = 6$, $n = 9$ and $L = [1, 6] \cup \{9\}$. The diagonal steps cross the corners, which are tree edges, and the arrows are the remaining tree edges; $T$ adds the leaves 7 and 8 at 9. The position $ba = 24$, marked $\circ$, lies in neither tree.

**Step 4: moving the remaining edges.** Let $M = \Gamma^+ \setminus T$, let $M_1$ be the positions in $M$ inside $L$, and let $M_2$ be the set of pairs $uv$ with $a < u < v < n$, together with $ba$. Then $M = M_1 \cup M_2$, and $ba \in M_1 \cap M_2$. Step 2 exchanges any two positions of $M_1$, and Lemma 5.6 on the component $[a, n]$ exchanges any two positions of $M_2 \setminus \{ba\}$, restoring all other edges. Suppose $ba \in \theta$ and $x \in M_2 \setminus \{ba\}$ is vacant. Let $d \ne c$ be an endpoint of $x$, and let $y$ be the edge joining $c$ and $d$. If $y \in \theta$, the component stays connected when we replace $cn$ by $x$, so Lemma 5.6 performs this replacement. Then Lemma 5.2 for the corners $bc$ and $an$ exchanges $ba$ for $cn$, and we obtain $\theta - ba + x$. If $y \notin \theta$, the same construction with $y$ in place of $x$ gives $\theta - ba + y$, and then we exchange $y$ and $x$. Reversing these paths exchanges in the other direction. Finally, a position of $M_1 \setminus M_2$ and one of $M_2 \setminus M_1$ are exchanged by routing through $ba$, as in Step 3 of Lemma 5.6. Thus all orientations containing $T$ with the same number of edges lie in one flip class, and with Lemma 5.1 this proves the lemma. ∎

Reversing the order of the vertices maps flips to flips, since it reverses the direction of all edges and preserves covers. Hence Lemma 5.9 also holds for the first two corners, with the graph induced on $\{1\} \cup [b', n]$ in place of $\Gamma[L]$ when the first two corners are $1a'$ and $b'c'$.

5.6. **The first vertex.** Suppose that the first two corners are $1a$ and $2b$, with $2 < a < b$. Let $\Gamma'$ be the graph induced on $2, \dots, n$. It has the corners of $\Gamma$ other than $1a$, and the same values of $r$ on $2, \dots, n$. Hence $\Gamma^+$ consists of $1a$, the set $(\Gamma')^+$ for $\Gamma'$, and the edges $1v$ for $v$ in

$$J = \{v : 1 < v < a,\ r(a) < r(v)\}.$$

For $v \in J$, let $vh_v$ be the corner leaving $v$. Then $h_v > a$, since a corner starting in $(1, a)$ does not end inside $[1, a]$, and $d_v = ah_v$ is a normal edge of $(\Gamma')^+$. Lemma 5.2 for the corners $1a$ and $vh_v$ exchanges $1v$ and $d_v$.

**Lemma 5.10.** *Suppose that Theorem 4.9 holds for $\Gamma'$, and that $(\Gamma')^+ \setminus \{d_v\}$ is connected for every $v \in J$. Then Theorem 4.9 holds for $\Gamma$.*

*Proof.* Let $\theta \subseteq \Gamma^+$ be connected. If the edges of $\theta$ inside $[2, n]$ do not form a connected graph, let $B$ be the component containing $a$. Some edge $1v \in \theta$ ends in another component, and then $v \in J$, the vertices $v$ and $h_v$ lie in the same component, and $d_v \notin \theta$. Lemma 5.2 replaces $1v$ by $d_v$, and the number of edges at 1 decreases. Hence we may assume that the edges inside $[2, n]$ form a connected graph.

Fix the set $I \subseteq J$ of vertices $v$ with $1v \in \theta$. Paths between vertices of $[2, n]$ avoid 1, so flips of $\Gamma'$ lift with the edges at 1 fixed, and by assumption all such orientations with the same $I$ and the same number of edges lie in one flip class.

Let $m$ be the number of edges of $(\Gamma')^+$, and let $s$ be the number of edges of $\theta$. The corners of $\Gamma'$ form a forest, which extends to a spanning tree of $(\Gamma')^+$, so connected orientations of $\Gamma'$ in $(\Gamma')^+$ exist with any number of edges between $n-2$ and $m$. Hence the possible sets are exactly the $I \subseteq J$ with

$$\max(0, s - 1 - m) \le |I| \le \min(|J|, s - 1 - (n-2)). \tag{9}$$

Suppose that $I$ and $I \setminus \{v\}$ both satisfy (9). The corresponding orientations of $\Gamma'$ have $t$ and $t+1$ edges, with $n-2 \le t < m$. Since $(\Gamma')^+ \setminus \{d_v\}$ is connected, there is a connected orientation of $\Gamma'$ in $(\Gamma')^+$ with $t$ edges avoiding $d_v$. Together with the edges $1w$, $w \in I$, Lemma 5.2 moves it to an orientation with the set $I \setminus \{v\}$. Thus the classes for $I$ and $I \setminus \{v\}$ agree.

Any two sets satisfying (9) are joined by such insertions and deletions: to exchange two elements we delete first if $|I|$ is above its lower bound, and insert first otherwise. This fails only if (9) allows a single cardinality and more than one set. If $J \neq \emptyset$, the hypothesis gives $m \ge n - 1$, and then a single allowed cardinality forces $I = \emptyset$ or $I = J$. Hence all connected orientations in $\Gamma^+$ with $s$ edges lie in one flip class, and Lemma 5.1 proves the lemma. ∎

5.7. **Proof of Theorem 4.9.** First let the natural graph be connected without articulations. We proceed by induction on $n$. If there is only one corner, the graph is complete, and Lemma 5.6 applies. If there are at most two corner chains, Lemma 5.5 applies. Otherwise let $1a$ and $b'c'$ be the first two corners. Since there are no articulations, consecutive maximal cliques share at least two vertices, so $1 < b' < a < c'$. If $b' > 2$, the reversed form of Lemma 5.9 reduces the claim to the graph induced on $\{1\} \cup [b', n]$, which has fewer vertices and no articulation. If

$b' = 2$, deleting the vertex 1 removes the corner $1a$ and gives a graph $\Gamma'$ without articulations, whose maximal cliques start with $[2, c']$. It has the same number of corner chains as $\Gamma$, at least three, so Lemma 5.8 for $\Gamma'$ verifies the hypothesis of Lemma 5.10, and induction applies to $\Gamma'$.

Finally, let the natural graph consist of the path $1, \dots, b$ followed by a graph without articulations on $[b, n]$. The path edges are corners, so they lie in every orientation we consider, and paths between vertices of $[b, n]$ do not use vertices below $b$. Hence restriction to $[b, n]$ is a bijection between the orientations in question of $\Gamma$ and of the graph induced on $[b, n]$, compatible with flips. This completes the proof of Theorem 4.9, and thereby of Theorem 2.11 for ordinary graphs.

## 6. Flip classes of circular graphs

In this section we prove Theorem 2.11 for circular graphs, by reduction to the ordinary case. The idea is to expand $\mathsf{K}(\theta)$ and $\mathsf{E}(\theta)$, for every admissible orientation, as the same signed sum over *balanced* sub-orientations, in which the lengths of the edges are differences of a potential. The coefficients turn out to be constant on flip classes, so the expansion can be grouped by flip classes. Finally, the orientations of a balanced flip class can be drawn on a line, where they form a union of ordinary flip classes.

Let $\Gamma$ be a circular decorated unit-interval graph. A strict edge is a corner, so it is never a short edge of a flip, and it is present in every orientation. Hence forgetting the strict markings, and including the strict edges in every orientation, identifies the flip classes of $\Gamma$ with flip classes of the undecorated graph, with the same graphs $\theta \cup S$. We therefore assume that $S = \emptyset$, and identify an orientation $\theta$ with the edge set of $\theta \cup S$. Throughout, orientations are admissible, and for a set of edges $Y$ we write $c(Y)$ for the number of connected components of the undirected graph on $\mathbb{Z}/n\mathbb{Z}$ with edges $Y$, isolated vertices included. Every edge $e$ has a *length* $\ell(e) \in \{1, \dots, n-1\}$.

We write $\mathsf{K}(\theta) = \mathrm{K}_\theta$, and $h_\theta$ for the whole map $z \mapsto \widetilde{\mathrm{hrv}}(\theta, z)$ on $\mathbb{Z}$. We work in the free abelian group on such maps, with basis elements $[h_\theta]$. The linear map $[h] \mapsto \mathrm{e}_\lambda$, where $\lambda$ is the partition of the fibre sizes of $h$ modulo $n$, sends $[h_\theta]$ to $\mathsf{E}(\theta)$.

### 6.1. Balanced orientations.

The signed expansion is indexed by subsets of an orientation of the following kind.

**Definition 6.1.** A set $F$ of edges is *balanced* if on each connected component of the undirected graph $(\mathbb{Z}/n\mathbb{Z}, F)$ there is a function $p$ to $\mathbb{Z}$ with $\ell(e) = p(v) - p(u)$ for every edge $e$ from $u$ to $v$ in $F$.

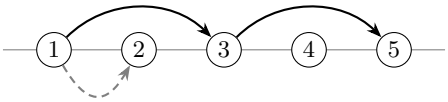
Equivalently, the lifted graph of $F$ is a disjoint union of translates of a graph on finitely many integers. Let $F \subseteq \theta$ be balanced, with such a potential $p$. Contract the connected components of $F$, keeping the edges of $\theta \setminus F$ as individual edges, possibly loops or parallel edges, and give the edge $e$ from $u$ to $v$ the *gain* $\delta(e) = \ell(e) - (p(v) - p(u))$. Changing $p$ by a constant on a component does not change the gain of a closed walk.

For a directed multigraph $Q$ with gains, let $\nu(Q) = 1$ if every directed cycle of $Q$ has negative gain, and $\nu(Q) = 0$ otherwise. Let $\zeta(Q) = 1$ if every edge of $Q$ lies on a directed cycle, and $\zeta(Q) = 0$ otherwise.

**Definition 6.2.** For a balanced $F \subseteq \theta$ with contracted graph $Q$, let $\chi(\theta, F) = (-1)^{c(F)-c(\theta)}\,\nu(Q)\,\zeta(Q)$. For $F$ not balanced, or not contained in $\theta$, we put $\chi(\theta, F) = 0$.

If $\theta$ itself is balanced, then $\chi(\theta, F) = [F = \theta]$. Indeed, restricting a potential of $\theta$ to $F \subsetneq \theta$ gives every edge of $\theta \setminus F$ gain zero, so any directed cycle containing one of them, a loop included, has gain zero.

**Example 6.3.** Let $\mathbf{a} = (2, 2, 2)$, so that $\Gamma$ is the complete circular graph on three vertices, and let $\theta$ consist of the edges $1 \to 2$, $1 \to 3$ and $3 \to 2$, of lengths 1, 2 and 2. It is not balanced: the path $1 \to 3 \to 2$ has length 4, while $1 \to 2$ has length 1. Lemma 6.4 below, applied to this cycle, gives $\mathsf{K}(\theta) = \mathsf{K}(F)$ and $[h_\theta] = [h_F]$ for $F = \{1 \to 3, 3 \to 2\}$. The set $F$ is balanced; its lift is the path $1 \to 3 \to 5$, so 1 and 3 have the highest reachable vertex $5 \equiv 2$, with or without the edge $1 \to 2$, and both values equal $\mathrm{e}_3$. In the lift below, the integer $z$ lies over the residue of $z$ modulo 3, the solid edges are the lift of $F$ starting in $\{1, 2, 3\}$, and the dashed edge is the lift of the deleted edge $1 \to 2$:




## 6.2. **Deleting edges of a cycle.**

**Lemma 6.4.** *Let $\theta$ be admissible, and let $Z$ be a cycle of the undirected graph of $\theta$, traversed so that its forward edges form a nonempty set $Z^+$, its backward edges form a set $Z^-$, and*

$$\sum_{e \in Z^+} \ell(e) > \sum_{e \in Z^-} \ell(e).$$

*Let $B = \theta \setminus Z^-$. Then*

$$\sum_{J \subseteq Z^-} (-1)^{|J|} \mathsf{K}(B \cup J) = 0 \qquad \text{and} \qquad \sum_{J \subseteq Z^-} (-1)^{|J|} [h_{B \cup J}] = 0.$$

*Proof.* Let $\kappa$ be a coloring which increases along the edges of $B$. The values of $\kappa$ telescope around $Z$, so $\sum_{Z^+}(\kappa(v) - \kappa(u)) = \sum_{Z^-}(\kappa(v) - \kappa(u))$, the sums ranging over the edges from $u$ to $v$. The left side is positive, so some edge of $Z^-$ increases. Toggling the first such edge, in a fixed order of $Z^-$, is a sign-reversing involution on the sets $J$ for which $\kappa$ increases along $B \cup J$, which proves the first identity.

For the second, fix $J$ and let $h = h_{B \cup J}$. The function $t(z) = h(z) - z$ is periodic, so it is a function on residues. For an edge from $u$ to $v$ in $B \cup J$, its lift ends at $u + \ell$ and $h(u) \geq h(u + \ell)$, so $t(u) - t(v) \geq \ell$. Telescoping $t$ around $Z$ gives

$$\sum_{Z^-}(t(u) - t(v)) = \sum_{Z^+}(t(u) - t(v)) \geq \sum_{Z^+} \ell > \sum_{Z^-} \ell.$$

Hence some edge of $Z^-$ satisfies $t(u) - t(v) > \ell$, that is, $h(u) > h(u + \ell)$ for its lifts. Adding this edge does not change $h$, by Lemma 3.1(i). Removing it does not change $h$ either: a path through a lift of this edge ends below $h(u + \ell) < h(u)$ and is not maximizing. Toggling the first edge of $Z^-$ with this property is therefore a sign-reversing involution which preserves $h$. ∎

If $\theta$ is not balanced, choose a spanning forest of $\theta$ and a potential on it. Some edge outside the forest violates the potential, and together with the path in the forest between its endpoints it forms a cycle with nonzero signed length. We traverse this cycle in the direction in which the forward edges are longer. Then $Z^- \neq \emptyset$, since otherwise $\theta$ would contain a directed cycle. Hence every set $B \cup J$ with $J \subsetneq Z^-$ in Lemma 6.4 has fewer edges than $\theta$.

6.3. **Expansion into balanced orientations.** An *ordered partition* of the vertex set of a directed multigraph $\Delta$ is a sequence $\omega = (\omega_1, \dots, \omega_k)$ of disjoint nonempty blocks covering it; it is *compatible* with $\Delta$ if every edge goes from some $\omega_i$ to some $\omega_j$ with $i \leq j$.

**Lemma 6.5.** *For every finite directed multigraph $\Delta$,*

$$\sum_{\omega} (-1)^{k(\omega)} = (-1)^{c(\Delta)}\, \zeta(\Delta),$$

*where $\omega$ ranges over the ordered partitions compatible with $\Delta$, and $k(\omega)$ is the number of blocks.*

*Proof.* A strongly connected component lies in one block of a compatible ordered partition, so we may contract the strong components and remove the loops. The edges of $\Delta$ all lie on directed cycles exactly when the resulting acyclic graph $\Delta'$ has no edges, and in that case $c(\Delta)$ is the number of vertices of $\Delta'$. We show by induction that the sum for $\Delta'$ is $(-1)^{|\Delta'|}$ if $\Delta'$ has no edges and 0 otherwise. The last block $U$ of a compatible ordered partition is a nonempty set closed under taking heads of edges, so the sum for $\Delta'$ is minus the sum over such $U$ of the sums for $\Delta' \setminus U$. By induction only the sets $U$ containing every head of an edge contribute, and the sum becomes $-(-1)^{|\Delta'|} \sum_U (-1)^{|U|}$ over the nonempty sets $U$ containing all heads. If $\Delta'$ has an edge, it has a source which is not a head, and the sum vanishes; otherwise it equals $-1$. ∎

**Proposition 6.6.** *For every admissible orientation $\theta$,*

$$\mathsf{K}(\theta) = \sum_{F \subseteq \theta} \chi(\theta, F)\, \mathsf{K}(F) \qquad \text{and} \qquad [h_\theta] = \sum_{F \subseteq \theta} \chi(\theta, F)\, [h_F].$$

*In particular $\mathsf{E}(\theta) = \sum_F \chi(\theta, F)\, \mathsf{E}(F)$.*

*Proof.* We proceed by induction on $|\theta|$. If $\theta$ is balanced, then $\chi(\theta, F) = [F = \theta]$. Otherwise choose a cycle as after Lemma 6.4. That lemma expresses both values of $\theta$ through the sets $B \cup J$ with $J \subsetneq Z^-$, and by induction it suffices to prove

$$\sum_{J \subseteq Z^-} (-1)^{|J|} \chi(B \cup J, F) = 0 \qquad \text{for all } F \subseteq \theta. \tag{10}$$

Fix $F$; we may assume that $F$ is balanced, with a potential $p$. Contract the components of $F$ and let $Q$ be the resulting multigraph with gains. Its edges are the edges of $B \setminus F$, which we call required, and the edges of $Z^-_F = Z^- \setminus F$, which we call optional. Only the sets $J$ containing $F \cap Z^-$ contribute. Writing $Q(J)$ for the graph of the required edges together with the optional edges in $J$, and using $c(B \cup J) = c(Q(J))$, the identity (10) becomes, up to a common sign,

$$\sum_{J \subseteq Z^-_F} (-1)^{|J| + c(Q(J))}\, \nu(Q(J))\, \zeta(Q(J)) = 0. \tag{11}$$

The cycle $Z$ projects to a closed walk in $Q$ which follows the required edges forwards and the optional edges backwards, and since the potential telescopes, its gain satisfies

$$\sum_{\text{required}} \delta - \sum_{\text{optional}} \delta = \sum_{Z^+} \ell - \sum_{Z^-} \ell > 0, \tag{12}$$

the sums on the left ranging over the edges of the walk.

By Lemma 6.5, the left side of (11) equals $\sum_\omega (-1)^{k(\omega)} \Sigma_\omega$, where $\Sigma_\omega$ is the sum of $(-1)^{|J|}$ over the $J$ such that $\omega$ is compatible with $Q(J)$ and $\nu(Q(J)) = 1$. Fix $\omega$, and assume that the required edges are compatible with $\omega$, since otherwise $\Sigma_\omega$ is empty. If an optional edge compatible with $\omega$ goes strictly upwards between blocks, it lies on no cycle of a graph compatible with $\omega$, and toggling it is a sign-reversing involution. Otherwise every compatible optional edge stays in one block. Along the closed walk, required edges do not decrease the block index, compatible optional edges traversed backwards keep it, and incompatible optional edges traversed backwards increase it strictly. The total change is zero, so the walk lies in a single block, and it remains to show that

$$\sum_J (-1)^{|J|}\, \nu(Q(J)) = 0, \tag{13}$$

the sum ranging over the sets $J$ of compatible optional edges.

Let $M$ exceed both the number of vertices and the number of optional edges of $Q$, and put $\gamma(e) = M\delta(e) + 1$. A simple cycle of length $r$ has $\gamma$-gain $Mg + r$, where $g$ is its gain, so $\nu(Q(J)) = 1$ if and only if no cycle has positive $\gamma$-gain. In that case let $\tau(y)$ be the maximum of 0 and the $\gamma$-gains of the directed walks ending at $y$. It is the least nonnegative function with $\tau(v) - \tau(u) \geq \gamma(e)$ for all edges $e$ from $u$ to $v$ of $Q(J)$. Summing $\tau(v) - \tau(u) - \gamma(e)$ around the closed walk, by (12) the optional edges exceed the required ones by $M(\sum \delta - \sum \delta) + (\#\text{required} - \#\text{optional}) > 0$, so some optional edge $e$ of the walk, from $u$ to $v$, has $\tau(v) - \tau(u) > \gamma(e)$. Adding such an edge preserves the inequalities, so $\tau$ does not change, and removing it does not change $\tau$ either, since a walk through it is not maximizing. Toggling the first such edge, in a fixed order of the optional edges, is therefore a sign-reversing involution, which proves (13) and the proposition. ∎

6.4. **Constancy on flip classes.** Let $C$ be a flip class of $\Gamma$, and for an admissible orientation $\eta$ put

$$a_C(\eta) = \sum_{\alpha \in C} \chi(\alpha, \eta).$$

**Lemma 6.7.** *The function $a_C$ is constant on every flip class.*

*Proof.* Let $\theta$ and $\eta$ differ by a flip which exchanges $ab$ for $bc$ across $ac$, in a window $a < b < c < a + n$ of the lift. Let $\alpha \in C$ with $\chi(\alpha, \theta) \neq 0$, so $\theta \subseteq \alpha$. Choose a potential $p$ of $\theta$ and contract the components of $\theta$. In the contracted graph, every directed cycle formed by edges of $\alpha \setminus \theta$ has negative gain, so with $\gamma = M\delta + 1$, where $M$ exceeds the number of vertices, there is a function $\tau$ on the contracted vertices with $\tau(v) - \tau(u) \geq \gamma(e)$ for these edges, as in the proof of Proposition 6.6. Then $z(v) = p(v) + \tau(v)/M$ satisfies $z(v) - z(u) = \ell(e)$ for the edges of $\theta$ and $z(v) - z(u) > \ell(e)$ for the edges of $\alpha \setminus \theta$. The lengths satisfy $\ell(ac) = \ell(ab) + \ell(bc)$, so $bc \notin \alpha$. Let $\alpha' = \alpha - ab + bc$.

A lifted path in $\alpha$ between the endpoints of $ab$ has total length $\ell(ab)$ and total increase of $z$ equal to $\ell(ab)$, so all its edges satisfy the equality, that is, they lie

in $\theta$. Since $ab$ and $ac$ are covers for $\theta$, they are covers for $\alpha$, and similarly $bc$ and $ac$ are covers for $\alpha'$, using $\eta$. The function $z$ increases along every edge of $\alpha'$, so $\alpha'$ is admissible, and $\alpha'$ differs from $\alpha$ by a flip; hence $\alpha' \in C$. The orientations $\theta$ and $\eta$ have the same components and the same potential, and $\alpha' \setminus \eta = \alpha \setminus \theta$, so $\chi(\alpha', \eta) = \chi(\alpha, \theta)$. The same construction from $\eta$ inverts $\alpha \mapsto \alpha'$, which proves $a_C(\theta) = a_C(\eta)$. ∎

Summing Proposition 6.6 over $C$ and grouping by flip classes $C'$, we obtain

$$\mathsf{K}(C) = \sum_{C'} a_C(C')\,\mathsf{K}(C'), \qquad \mathsf{E}(C) = \sum_{C'} a_C(C')\,\mathsf{E}(C'), \tag{14}$$

where $a_C(C')$ denotes the common value on $C'$, and $a_C(C') = 0$ unless $C'$ contains a balanced orientation.

### 6.5. **Balanced classes are ordinary.**

**Proposition 6.8.** *Let $C$ be a flip class containing a balanced orientation. Then $\mathsf{K}(C) = \mathsf{E}(C)$.*

*Proof.* Let $p$ be a potential of a balanced orientation in $C$. On each component, $p(v) - v$ is constant modulo $n$, so changing $p$ by constants we may assume that $p(v) \equiv v \pmod n$ and that $p$ is injective. The relation $\ell(ac) = \ell(ab) + \ell(bc)$ shows that $p$ is a potential for every orientation in $C$. Order the vertices by the values of $p$, and let $\Gamma'$ be the ordinary graph on these ranks in which a pair is a natural edge if the corresponding lifts $p(u) < p(v)$ form a lifted natural edge. It is a unit-interval graph since the lifted natural edges are closed under subintervals.

An orientation $\xi$ of $\Gamma'$ maps to the circular orientation consisting of the residues of its lifted edges. A lifted path starting at $p(u)$ only visits the chosen lifts $p(w)$, since $p$ is injective modulo $n$. Hence the image is admissible, its covers correspond to those of $P_+(\xi)$, its colorings correspond to those of $\xi$, and its highest reachable lifts are the values $p(\mathrm{hrv}(\xi, u))$. So the map preserves $\mathsf{K}$ and $\mathsf{E}$, and maps ordinary flips to circular flips; the window has length less than $n$ since the long edge is natural. Every orientation in $C$ is the image of some $\xi$, since $p$ is a potential for it. Hence the set of $\xi$ whose image lies in $C$ is closed under ordinary flips, and is a union of ordinary flip classes, on which the ordinary case of Theorem 2.11 gives $\mathsf{K} = \mathsf{E}$. ∎

*Proof of Theorem 2.11 for circular graphs.* By (14) and Proposition 6.8, $\mathsf{K}(C) = \sum_{C'} a_C(C')\mathsf{K}(C') = \sum_{C'} a_C(C')\mathsf{E}(C') = \mathsf{E}(C)$. ∎

## 7. Consequences

### 7.1. **The elementary expansion.**

*Proof of Corollary 2.12.* By Lemma 2.2, and since asc is constant on flip classes,

$$\mathrm{LLT}_\Gamma(\mathbf{x}; q+1) = \sum_\theta q^{\mathrm{asc}(\theta)} \mathrm{K}_\theta(\mathbf{x}) = \sum_C q^{\mathrm{asc}(C)} \mathsf{K}(C),$$

the second sum ranging over the flip classes. By Theorem 2.11 this equals

$$\sum_C q^{\mathrm{asc}(C)} \mathsf{E}(C) = \sum_\theta q^{\mathrm{asc}(\theta)} \mathrm{e}_{\lambda(\theta)}(\mathbf{x}).$$ ∎

Thus the coefficient of $q^k \mathrm{e}_\lambda$ in $\mathrm{LLT}_\Gamma(\mathbf{x}; q+1)$ is the number of admissible orientations $\theta$ with $\mathrm{asc}(\theta) = k$ and $\lambda(\theta) = \lambda$, and every flip class contributes a sum of such terms. Substituting $q-1$ for $q$, we also have $\mathrm{LLT}_\Gamma(\mathbf{x}; q) = \sum_\theta (q-1)^{\mathrm{asc}(\theta)} \mathrm{e}_{\lambda(\theta)}(\mathbf{x})$, which is the form in which the expansion was stated in [AS22]. For ordinary graphs, Corollary 2.12 is Theorem 2.4, and the proof above does not use the recursion of [AS22]. For circular graphs it answers the question of [AS22, Sec. 6.8].

7.2. **LLT cumulants.** We now refine the elementary expansion of LLT cumulants of M. Dołęga and M. Kowalski [DK22] to flip classes, for circular graphs. Let $\Gamma$ be a circular decorated unit-interval graph and $U$ a nonempty set of residues. List the lifts of the elements of $U$ increasingly as $(u_j)_{j\in\mathbb{Z}}$, so that $u_{j+m} = u_j + n$ with $m = |U|$, and put $b'(j) = \min\{k : u_k \geq b(u_j)\}$. Then $b'$ is weakly increasing, $j - m < b'(j) \leq j$, and $b'(j+m) = b'(j) + m$, so it defines a circular unit-interval graph on $U$ whose lifted natural edges are the pairs $u_k u_j$ with $b(u_j) \leq u_k < u_j$. A strict edge $u_k u_j$ of $\Gamma$ remains a corner, since $b(u_j) = u_k < b(u_j + 1)$ gives $b'(j) = k < b'(j+1)$. We call the resulting graph the *induced* graph $\Gamma[U]$. Its natural edges are exactly the natural edges of $\Gamma$ with both endpoints in $U$, and its strict edges are the strict edges of $\Gamma$ inside $U$.

Let $r \geq 1$, and let $f : \mathbb{Z}/n\mathbb{Z} \to [r]$ be surjective and constant on the endpoints of every strict edge. For nonempty $B \subseteq [r]$ write $\Gamma[B]$ for the graph induced on $f^{-1}(B)$. Let $\Pi_r$ be the lattice of set partitions of $[r]$, with maximum $\hat{1} = \{[r]\}$. Following [DK22], the *LLT cumulant* is

$$\kappa_{\Gamma,f}(\mathbf{x}; q) \coloneqq (q-1)^{1-r} \sum_{\pi \in \Pi_r} (-1)^{|\pi|-1} (|\pi|-1)! \prod_{B \in \pi} \mathrm{LLT}_{\Gamma[B]}(\mathbf{x}; q).$$

For an admissible orientation $\theta$, let $\sigma_f(\theta)$ be the set partition of $[r]$ into the connected components of the undirected multigraph on $[r]$ with an edge $f(u)f(v)$ for every edge $uv$ of $\theta \cup S$. We call $\theta$ *$f$-connected* if $\sigma_f(\theta) = \hat{1}$.

**Proposition 7.1** (cf. [DK22, Thm. 3.8])**.** *We have*

$$\kappa_{\Gamma,f}(\mathbf{x}; q+1) = \sum_\theta q^{\mathrm{asc}(\theta)+1-r} \, \mathrm{K}_\theta(\mathbf{x}),$$

*the sum ranging over the $f$-connected admissible orientations of $\Gamma$, and all exponents are nonnegative.*

*Proof.* For $\pi \in \Pi_r$, the orientations of $\Gamma$ without ascending edges between different blocks of $\pi$ are the tuples of orientations of the graphs $\Gamma[B]$, $B \in \pi$, since no strict edge joins different blocks. By Lemma 2.2 for each $\Gamma[B]$,

$$\prod_{B \in \pi} \mathrm{LLT}_{\Gamma[B]}(\mathbf{x}; q+1) = \sum_{\sigma_f(\theta) \leq \pi} q^{\mathrm{asc}(\theta)} \mathrm{K}_\theta(\mathbf{x}).$$

Inserting this in the definition, the coefficient of a fixed $\theta$ is $q^{\mathrm{asc}(\theta)+1-r}$ times $\sum_{\sigma_f(\theta) \leq \pi} (-1)^{|\pi|-1}(|\pi|-1)!$. Since $(-1)^{|\pi|-1}(|\pi|-1)!$ is the Möbius function of $\Pi_r$ from $\pi$ to $\hat{1}$, this sum is $[\sigma_f(\theta) = \hat{1}]$. An $f$-connected orientation has at least $r-1$ ascending edges, since strict edges become loops. ∎

By Lemma 2.8, being $f$-connected is constant on flip classes. Grouping Proposition 7.1 by flip classes and applying Theorem 2.11, we obtain the following.

**Corollary 7.2.** *With the notation above,*

$$\kappa_{\Gamma,f}(\mathbf{x}; q+1) = \sum_{C} q^{\mathrm{asc}(C)+1-r} \sum_{\theta \in C} \mathrm{e}_{\lambda(\theta)}(\mathbf{x}),$$

*where $C$ ranges over the flip classes of $f$-connected admissible orientations. In particular, every flip class contributes an elementary-positive summand.*

For ordinary graphs, summing over the classes recovers the elementary expansion of [DK22, Thm. 3.17].

7.3. **Questions.** For a unit-interval graph without strict edges, the chromatic quasisymmetric function and the unicellular LLT polynomial are related by a plethysm, see [AS22, Lem. 6.17], and B.-H. Hwang conjectured that the flip classes of acyclic orientations contribute elementary-positive summands to the chromatic quasisymmetric function [Hwa24, Conj. 3.10]. The plethystic relation does not preserve elementary positivity term by term: already $(q-1)^{-1}\mathrm{e}_2[(q-1)\mathbf{x}] = (q+1)\mathrm{e}_2 - \mathrm{e}_{11}$.

**Question 7.3.** Is there a relation between Hwang's flip classes of acyclic orientations and the flip classes of Definition 2.5 which explains both positivity phenomena?

**Question 7.4.** Give a bijective proof of Theorem 2.11, that is, a bijection between the strict $P_+(\theta)$-partitions for $\theta \in C$ and the strict partitions of the chains of the highest reachable fibres for $\theta \in C$, preserving the multiset of colors.

**Acknowledgements.** The first named author thanks Robin Sulzgruber for the initial discussions about flips.

**AI assistance.** We used OpenAI Codex and Claude for proof development, Lean formalization, and preparation of the text and diagrams. Theorem A and the elementary expansions have been verified in Lean. The authors are responsible for the final mathematical content and exposition.

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